

Extremality shift of rotating black holes at finite Gauss–Bonnet coupling

Abstract: Higher-curvature terms move the extremality bound of a black hole, and how far they move it once the coupling is no longer small is not something first-order perturbation theory can say. In five dimensions with two equal angular momenta, the Gauss–Bonnet shift of the extremal mass is known only to that first order in the coupling α [1]; the extremal solutions built numerically at finite coupling [2] map the domain of existence, not $M_{\text{ext}}(J, \alpha)$ or its slope. We construct the equal-spin family at finite coupling and measure the thermodynamic potential Ψ conjugate to α on each solution, by differentiating the discretised field equations implicitly; in the zero-temperature limit the extended first law identifies it with $\partial M_{\text{ext}}/\partial \alpha|_J$. On the branch connected to Myers–Perry that slope never changes sign. It falls from its perturbative value π to a minimum of 1.07 at intermediate coupling, then turns around and rises again, and it is still positive at the largest coupling we reach, $y = \alpha/J^{2/3} = 0.511$. The extremal mass therefore grows throughout, ending some 31% above its Einstein value, so the bound tightens rather than relaxes; by that point the first-order formula overshoots the shift by a factor of 1.32. Our calculation reproduces the static solution in closed form, the perturbative potential found in the literature, the Smarr relation, the on-shell Gauss–Bonnet action and the extremal near-horizon entropy.

Keywords: Black Holes, Classical Theories of Gravity, Higher Curvature Gravity

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1 Introduction

In Einstein gravity a rotating black hole cannot carry arbitrary angular momentum at a given mass. For Kerr the bound is $M^2 \geq J$; for the five-dimensional Myers–Perry solution [3] with two equal angular momenta $J_1 = J_2 = J$ it reads $M \geq \frac{3}{2}\pi^{1/3}J^{2/3}$. The black hole that saturates such a bound is extremal: its temperature vanishes and its horizon becomes degenerate. The bound is a property of the field equations rather than of black holes in general, and adding higher-curvature terms to the action moves it. How far, and in which direction, is what we compute here.

Two lines of work make the sign of that displacement interesting. The weak gravity conjecture [4, 5] reads the shift of an extremality bound as a statement about which states a black hole may decay into, and the entropy–extremality relation of Goon and Penco [6] ties the same shift to the sign of the entropy correction of nearby non-extremal states. Both are usually stated for *charged* black holes, where $M \geq M_{\text{ext}}(Q)$ compares a mass with a gauge charge. The family studied here is uncharged and carries angular momentum instead; angular momentum is not a charge under a gauge field, there is no corresponding light state to appeal to, and the spectrum arguments do not transfer without extra assumptions that we do not make. We therefore use the Goon–Penco relation only for what it is in this setting, an identity of the extended first law relating $\partial S/\partial\alpha|_{M,J}$ to $\partial M_{\text{ext}}/\partial\alpha|_J$, and draw no weak-gravity consequence from it.

The correction we study is the Gauss–Bonnet term, the unique quadratic curvature invariant whose field equations remain of second order [7] and the first nontrivial Lovelock correction in five dimensions. No rotating black hole of this theory is known in closed form, and the literature reaches finite coupling by two different routes. The first is perturbative in α . Reall and Santos [8] showed that the $\mathcal{O}(\alpha)$ thermodynamics of a black hole can be obtained without ever solving for the corrected metric; the corresponding results for Myers–Perry black holes in a general quadratic-curvature theory are given in [9] and the corrected metric itself in [10]. For the equal-spin family, Ma, Li and Lü [1] obtained the extremal mass at first order,

$$M_{\text{ext}} = \frac{3}{2}\pi^{1/3}J^{2/3} + \pi\alpha + \mathcal{O}(\alpha^2), \quad (1)$$

with J the angular momentum in each plane. The initial slope is positive, and (1) says nothing about the slope once α is no longer infinitesimal.

The second route is numerical. Imposing $J_1 = J_2$ enhances the symmetry enough to factor out the angular dependence, and the field equations reduce to ordinary differential equations in the radius. Brihaye, Kleihaus, Kunz and Radu [11] used that reduction to construct the non-extremal equal-spin family at finite coupling, and obtained the entropy of its extremal limit algebraically from the near-horizon geometry with Sen’s entropy function [12]. Kleihaus, Kunz and Radu [2] extended the scan and, instead of extrapolating in temperature, built the extremal solutions *directly*, in a radial gauge that places the degenerate horizon at the origin, and computed their mass, angular momentum, area and entropy. With those solutions they mapped the domain of existence of the family: the Einstein bound $j \leq 1$ on the scaled spin survives at finite coupling, the static mass bound $M > 3\pi\alpha/4$ holds for the rotating solutions as well, and the extremal set is conjectured to terminate at the nakedly singular static configuration with $M = 3\pi\alpha/4$ and $J = S = 0$. The last of these is a conjecture rather than a numerical result: section 3.2 of [2] records that the limit could not be reached, extrapolates the extremal curves of its figure 1 to the point $M = \frac{3}{4}\pi\alpha$, $J = A_H = S = 0$ by hand (the dotted line there), and supports the extrapolation with the exact rotating squashed $\text{AdS}_2 \times S^3$ near-horizon solution of [11] rather than with solutions constructed at the endpoint.

What that leaves open is the function the perturbative result (1) is an expansion of. Scale invariance makes the extremal branch a single curve in the reduced variables of [2], and that curve already fixes $M_{\text{ext}}(J, \alpha)$: with $\mu = M_{\text{ext}}/J^{2/3}$ and $y = \alpha/J^{2/3}$, the definitions in section 2.3 give $\mu = \mu(0)j^{-2/3}$ and $y = 4x\mu/(3\pi)$, so the domain of existence reported there contains the

extremal mass at fixed angular momentum. What it does not contain is the derivative. That function is not reported, and a slope read off a plotted domain boundary is not a quantity one would set beside the coefficient π of (1); nor is differencing the masses of separately constructed extremal solutions, since each carries its own construction error and the difference is taken between neighbouring couplings. The derivative is what the rest of this paper measures, and it is measured rather than differenced.

We take the derivative directly instead. The extended first law of Lovelock gravity [13, 14] promotes α to a thermodynamic variable with a conjugate potential $\Psi = \partial M / \partial \alpha|_{S,J}$, which we measure on each rotating solution by differentiating the discretised field equations implicitly — one linear solve, no step size, no second solution. On a regular extremal branch the first law identifies $\lim_{T \rightarrow 0} \Psi$ with $\partial M_{\text{ext}} / \partial \alpha|_J$, so the slope is obtained pointwise at each coupling rather than reconstructed from a difference of masses; integrating it then returns the mass curve itself, and the two routes can be compared. We follow the branch continuously connected to Myers–Perry at 13 couplings, from $\alpha = 0$ to $y = \alpha / J^{2/3} = 0.511$.

The slope is positive throughout the sampled range, so the extremal mass at fixed angular momentum increases with the coupling and the bound on the scaled spin tightens. It is not, however, constant: it falls well below the perturbative value π , reaching 1.07 near $y = 0.175$, and then rises again to 1.44. Figure 4 shows the resulting extremal mass, together with the first-order line (1) it departs from and the same branch drawn in the domain-of-existence plane of [2]. Section 6.1 reads the three features — the positive sign, the minimum of the slope and the subsequent rise — in terms of the scaled spin, the horizon angular velocity and the mass bound of [2].

It is worth separating what is new here from what is recovered. New: the Gauss–Bonnet potential Ψ measured on rotating solutions at finite coupling, across the whole family and not only at its extremal end, together with the sign change it undergoes; the extremal slope $\partial M_{\text{ext}} / \partial \alpha|_J$ obtained pointwise at 13 couplings, with its minimum and subsequent rise resolved; the extremal mass curve that slope integrates to, and the size of its departure from (1); the reading of Ψ as an on-shell Gauss–Bonnet action, tested against the response; and the identification of the slope minimum with the maximum of $\Omega_H J^{1/3}$ on the extremal branch. Recovered rather than new: the equal-spin family of [11] and its profiles; the closed-form static solution; the $\mathcal{O}(\alpha)$ potential implied by [9] and the first-order extremal mass π of [1]; the near-horizon extremal entropy of [11]; and the extremal branch itself, which [2] constructs directly and we reach by extrapolation in temperature, together with its bounds $j \leq 1$ and $x < 1$. One quantity from the literature that we do not reproduce, an ergosurface radius of [11], is reported in section 6.

Throughout, “the extremal branch” means the zero-temperature endpoint of the equal-spin family continuously connected to Myers–Perry, and nothing wider. Several distinct quantities are all called “spin” in this subject, and which one is held fixed changes the statement; they are separated in table 1 together with the conversions to the conventions of the papers we compare with.

Section 2 specifies the solutions and observables. Section 3 defines the potential and explains how it is evaluated. Results follow in section 4 and the comparisons with known results in section 5; section 6 interprets the result and states its limitations, and appendix A describes the numerical construction, the uncertainty budget and the sensitivity of the extrapolation.

2 The equal-spin family and its thermodynamics

2.1 Action and ansatz

We use

$$I = \frac{1}{16\pi G} \int d^5x \sqrt{-g} (R + \alpha \mathcal{L}_{\text{GB}}), \quad \mathcal{L}_{\text{GB}} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \quad (2)$$

and set $G = 1$. The coupling has dimensions of length squared. We restrict the study to $\alpha \geq 0$ and the branch connected continuously to Einstein gravity. This sign choice is a restriction of the study, not a consequence of the graviton propagator about flat space: the Gauss–Bonnet contribution to the field equations is quadratic in curvature and vanishes at linear order about Minkowski space. The equations are $G_{\mu\nu} + \alpha H_{\mu\nu} = 0$, with $H_{\mu\nu}$ the Lanczos tensor, and contain at most second derivatives [7].

The equal-spin ansatz [11] is

$$ds^2 = -b(r)dt^2 + \frac{dr^2}{f(r)} + r^2[d\theta^2 + \sin^2\theta \cos^2\theta (d\varphi_1 - d\varphi_2)^2] \\ + h(r)[\sin^2\theta d\varphi_1 + \cos^2\theta d\varphi_2 - w(r)dt]^2. \quad (3)$$

Here $\theta \in [0, \pi/2]$ and $\varphi_i \in [0, 2\pi)$. The four radial functions b, f, h, w are unknown. The horizon is at $r = r_H$, where $b = f = 0$, and its angular velocity is $\Omega_H = w(r_H)$. Asymptotic flatness requires $b, f, h/r^2 \rightarrow 1$ and $w \rightarrow 0$. Regularity and these boundary conditions define the numerical problem. The solutions are stationary: their metric is independent of time, although their horizons rotate.

We set $r_H = 1$ in the numerical construction. The horizon angular velocity Ω_H has dimensions of inverse length, so the family is labelled by the dimensionless combination

$$q = r_H \Omega_H, \quad (4)$$

which is what survives the rescaling of section 2.3. In the solver’s units, $r_H = 1$, the numerical values of q and Ω_H coincide; they are nevertheless different quantities, and only q labels a solution independently of the chosen length unit. All spins quoted below are values of q .

The range of q is fixed at $\alpha = 0$ by Myers–Perry. Write that solution in Boyer–Lindquist form, with rotation parameter a and horizon at $\rho = r_+$. The ansatz (3) has $g_{\theta\theta} = r^2$ whereas Boyer–Lindquist has $g_{\theta\theta} = \rho^2 + a^2$, so the two radial coordinates are related by $r^2 = \rho^2 + a^2$, and in particular $r_H^2 = r_+^2 + a^2$. Using $\Omega_H = a/(r_+^2 + a^2)$,

$$q = r_H \Omega_H = \frac{a}{r_H}, \quad u \equiv \frac{a^2}{r_+^2} = \frac{q^2}{1 - q^2}. \quad (5)$$

Equal-spin Myers–Perry is extremal at $r_+ = a$, where $r_H^2 = 2a^2$; hence q runs over $[0, 1/\sqrt{2})$ on the vacuum family and reaches $1/\sqrt{2}$ exactly at extremality. That value is a property of the Einstein solution and not a definition: at finite coupling the extremal q is an output of the construction, and the solutions reported below reach $q > 1/\sqrt{2}$. The tabulated numerical α is α/r_H^2 in these units; statements about the physical coupling are made in the scale-invariant variables of section 2.3.

2.2 Charges, entropy and the first law

The asymptotic coefficients determine the mass and each angular momentum:

$$b = 1 + \frac{U}{r^2} + \mathcal{O}(r^{-4}), \quad w = \frac{W}{r^4} + \mathcal{O}(r^{-6}), \quad M = -\frac{3\pi}{8}U, \quad J = \frac{\pi}{4}W. \quad (6)$$

The normalisations agree with [11]. The horizon temperature is $T = \sqrt{b'(r_H)f'(r_H)}/(4\pi)$. We use Wald entropy [15, 16], which for this stationary Lovelock horizon has the Jacobson–Myers form [17]:

$$S = \frac{1}{4} \int_{\mathcal{H}} d^3x \sqrt{\sigma} (1 + 2\alpha \tilde{R}), \quad \tilde{R} = \frac{8}{r_H^2} - \frac{2h(r_H)}{r_H^4}. \quad (7)$$

Thus S depends on α both explicitly and through the geometry. The intrinsic expression applies to the stationary horizon cross-section; we do not use it as an entropy prescription for dynamical horizons.

Promoting the coupling to a thermodynamic variable gives the extended first law of Lovelock gravity [13, 14], the second of which treats stationary black holes explicitly:

$$dM = T dS + 2\Omega_H dJ + \Psi d\alpha. \quad (8)$$

The two equal angular momenta contribute $\Omega_1 dJ_1 + \Omega_2 dJ_2 = 2\Omega_H dJ$ here. The potential Ψ is a thermodynamic response coefficient of a whole solution and not an additional radial field: it measures the change of mass per unit change of coupling at fixed entropy and angular momenta. Varying α compares equilibrium solutions of theories with different couplings; it does not describe a time-dependent change of the coupling within one spacetime. Applying Euler’s theorem to the scaling weights listed in section 2.3 — M and α of weight two, S and each J_i of weight three — yields

$$2M = 3TS + 6\Omega_H J + 2\alpha\Psi. \quad (9)$$

The integers here are scaling weights and nothing else. The 3 multiplies TS because S has weight three, the 2 multiplies $\alpha\Psi$ because α has weight two, and the 6 is 3×2 : the weight of J times the number of equal spins. Only that last factor of two reappears in the differential relations (12) and (13) below. The weights 3 and 2 do not, and their absence there is not an inconsistency: (9) is a finite algebraic relation among the charges of a *single* solution, obtained from the scaling, whereas (8) and everything derived from it relates *neighbouring* solutions and knows nothing about the scaling. Neither statement implies the other, and the two sets of coefficients should not be expected to match. We evaluate (9) as an internal consistency check. For $\alpha \neq 0$ it also gives $\Psi = (2M - 3TS - 6\Omega_H J)/(2\alpha)$, but subtraction in its numerator is ill-conditioned as $\alpha \rightarrow 0$. At $\alpha = 0$ it cannot determine Ψ . The response calculation below avoids this division.

2.3 Scale-invariant variables

Under $r_H \rightarrow \lambda r_H$, the quantities (M, α) scale as λ^2 , (S, J) as λ^3 and (T, Ω_H) as λ^{-1} . The potential Ψ is dimensionless. For nonzero spin we define

$$y = \frac{\alpha}{J^{2/3}}, \quad \tau = TJ^{1/3}, \quad \mu = \frac{M}{J^{2/3}}, \quad \sigma = \frac{S}{J}. \quad (10)$$

On the extremal branch, scale invariance implies $M_{\text{ext}}(J, \alpha) = J^{2/3}\mu(y)$, hence $\partial M_{\text{ext}}/\partial \alpha|_J = \mu'(y)$. This rescaling lets solutions computed at fixed r_H describe the extremal mass at fixed J .

Since τ vanishes both at extremality and as $J \rightarrow 0$ in the static limit, it is unsuitable as a single coordinate for the whole family. For the non-extremal plots we use [2]

$$x = \frac{3\pi\alpha}{4M}, \quad j = \left(\frac{3}{2}\right)^{3/2} \sqrt{\pi} \frac{J}{M^{3/2}}, \quad t_H = 4\sqrt{\frac{2\pi}{3}} T\sqrt{M}. \quad (11)$$

The normalisation gives $t_H = 1$ for the static Einstein solution and $j = 1$ for extremal Myers–Perry. Both are properties of the normalisation, not results, so recovering them is a check of the numerics rather than of the physics: the value measured on our own extrapolated Einstein extremal state is $j = 1.00000000$, which tests the mass and spin extraction together with the zero-temperature extrapolation. At finite coupling the static temperature is instead $t_H = \sqrt{1 + 2\bar{\alpha}}/(1 + 4\bar{\alpha})$, where $\bar{\alpha} = \alpha/r_H^2$; it is not unity.

quantity	here	elsewhere
angular momentum	J is the momentum in <i>each</i> plane, $J_1 = J_2 = J$	the total is $2J$; the charge normalisations of (6) are those of [11], and j , x and σ below are all formed from the individual J
radial coordinate	r , fixed by $g_{\theta\theta} = r^2$ in (3); horizon at r_H	Boyer–Lindquist ρ , horizon at r_+ , with $r^2 = \rho^2 + a^2$ and $r_H^2 = r_+^2 + a^2$
spin label	$q = r_H \Omega_H$, dimensionless; numerically equal to Ω_H only because we set $r_H = 1$	rotation parameter $a = q r_H$, and $u = a^2/r_+^2 = q^2/(1 - q^2)$; the vacuum family is extremal at $q = 1/\sqrt{2}$
scaled spin	$j = (3/2)^{3/2} \sqrt{\pi} J/M^{3/2}$, normalised to $j = 1$ on extremal Myers–Perry	the same j as [2], whose domain bound is $j \leq 1$
Gauss–Bonnet coupling	α multiplying $\mathcal{L}_{\text{GB}}$ inside $(16\pi G)^{-1}$ in (2); tables print α/r_H^2	$\alpha_{\text{ref.}} = 4\alpha$ in [11]; the coefficient of $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ in [9] equals α on a Ricci-flat background
scale-invariant coupling	$y = \alpha/J^{2/3}$, used wherever a statement is about the physical coupling	$x = 3\pi\alpha/(4M)$ of [2], whose domain bound is $x < 1$

Table 1. Every conversion used in this paper. The distinction that matters most for the result is the last but one: q labels a solution at fixed horizon radius, j labels it at fixed mass, and y is what is held fixed when the extremal mass is differentiated with respect to the coupling at fixed J .

2.4 Conventions and conversions

Five quantities in this subject are routinely called the spin, the radius or the coupling, and papers differ on which one they print. Table 1 collects every conversion used below in one place, so that no comparison with the literature has to be reconstructed from the surrounding text. It is a statement of conventions and contains no measured number.

3 What the Gauss–Bonnet potential measures

3.1 Definition and evaluation along the numerical family

By definition, Ψ is the derivative of mass with respect to α at fixed entropy and both angular momenta. The solver, however, fixes r_H and Ω_H . Along its coupling direction, both entropy and angular momentum generally change, so the derivative the solver delivers is not the one wanted. Let $\alpha \mapsto (M(\alpha), S(\alpha), J(\alpha))$ be the one-parameter family of solutions at fixed (r_H, Ω_H) . Dividing (8) by $d\alpha$ along that family and solving for Ψ gives

$$\Psi = \left. \frac{\partial M}{\partial \alpha} \right|_{S,J} = \left. \frac{\partial M}{\partial \alpha} \right|_{r_H, \Omega_H} - T \left. \frac{\partial S}{\partial \alpha} \right|_{r_H, \Omega_H} - 2\Omega_H \left. \frac{\partial J}{\partial \alpha} \right|_{r_H, \Omega_H}. \quad (12)$$

The subtraction removes the mass changes associated with entropy and spin, leaving the response to the coupling itself. Equation (12) is therefore valid for the rotating stationary solutions, not only for the static member. It does not equate partial derivatives with different constraints: it equates one constrained derivative to a combination of three derivatives evaluated along a different path.

The single factor of two in (12) counts the two equal spins and nothing more. The weights 3 and 2 of the Smarr relation (9) are absent for the reason given there: (12) is the differential first

law divided by $d\alpha$, not the scaling relation differentiated.

Nor do terms such as $S \partial_\alpha T$ or $J \partial_\alpha \Omega_H$ appear. Such terms would arise from differentiating the finite expression for M contained in (9), in which T , S , Ω_H , J and Ψ all depend on α . We do not take that route. In (8) the intensive quantities T and Ω_H multiply the variations dS and dJ and are not themselves varied, which is why (12) has exactly three terms. Differentiating (9) and eliminating dM with (8) yields instead the Gibbs–Duhem-type constraint $T dS + 3S dT + 2\Omega_H dJ + 6J d\Omega_H + 2\alpha d\Psi = 0$, which the solutions satisfy but which determines nothing new. More generally any differentiable path through equilibrium solutions with $d\alpha \neq 0$ gives the same Ψ after subtracting the entropy and spin contributions and dividing by $d\alpha$.

3.2 Connection with extremality

Suppose the sampled family approaches a regular extremal branch, on which the entropy is a function $S_{\text{ext}}(J, \alpha)$ of the remaining charges. The extremal mass is then the composition $M_{\text{ext}}(J, \alpha) = M(S_{\text{ext}}(J, \alpha), J, \alpha)$, and the chain rule together with (8) gives

$$\left. \frac{\partial M_{\text{ext}}}{\partial \alpha} \right|_J = T \left. \frac{\partial S_{\text{ext}}}{\partial \alpha} \right|_J + \left. \frac{\partial M}{\partial \alpha} \right|_{S,J} = T \left. \frac{\partial S_{\text{ext}}}{\partial \alpha} \right|_J + \Psi. \quad (13)$$

No Ω_H term appears because J is held fixed, and no scaling weights appear because this is a chain rule and not a form of (9). Since T vanishes identically along the extremal branch, the first term drops as soon as $\partial S_{\text{ext}}/\partial \alpha|_J$ is finite, leaving $\partial M_{\text{ext}}/\partial \alpha|_J = \Psi_{\text{ext}}$, where $\Psi_{\text{ext}} \equiv \lim_{T \rightarrow 0} \Psi$ along the branch.

To infer this value from non-extremal numerical solutions we assume that their potentials have a continuous limit on that branch and that the entropy term has a vanishing limit. A bounded extremal entropy derivative is a sufficient regularity condition. Agreement with the algebraic near-horizon entropy and smooth sampled data supports this assumption over the range studied, but a finite table is not a proof of it.

At positive temperature the same first law implies the exact identity

$$\left. \frac{\partial S}{\partial \alpha} \right|_{M,J} = -\frac{\Psi}{T}.$$

Combining it with (13) gives

$$\left. \frac{\partial M_{\text{ext}}}{\partial \alpha} \right|_J = \lim_{T \rightarrow 0} \left(-T \left. \frac{\partial S}{\partial \alpha} \right|_{M,J} \right),$$

which is the entropy–extremality relation of Goon and Penco [6] for this family: the correction to the extremal mass is recovered from the correction to the entropy of nearby non-extremal states. We use it in that direction only, with α in the role their relation gives to the coefficient of a higher-derivative term and J in the role it gives to the conserved charges. In particular, positive Ψ means decreasing entropy at fixed mass and angular momenta, which is not the same as the entropy along the extremal branch, where the mass changes with α as well.

3.3 Numerical response by implicit differentiation

Equation (12) needs three derivatives of a numerically constructed solution with respect to the coupling. The obvious way to obtain them is to solve the boundary-value problem at α and at $\alpha \pm \Delta\alpha$ and difference the resulting charges. We avoid this. Such a difference carries a truncation error growing like $\Delta\alpha^2$ that must be traded against a cancellation error growing like the solver tolerance divided by $\Delta\alpha$, and the trade-off worsens near extremality, where the solutions are hardest to converge. The procedure below has no step size.

Let $u \in \mathbb{R}^{4N}$ collect the four metric amplitudes at the N collocation nodes, and write the discretised interior equations together with the horizon and asymptotic conditions as

$$R_N(u; \alpha, r_H, \Omega_H) = 0,$$

which is $4N$ equations for $4N$ unknowns. A converged run returns a root $u(\alpha)$ at fixed (r_H, Ω_H) . The useful observation is that R_N does not merely vanish at one point: it vanishes *identically* in α along the branch of roots. Its *total* derivative in α therefore vanishes too, and that total derivative expands by the chain rule into a term through u and a term through the explicit α :

$$\frac{d}{d\alpha} R_N(u(\alpha); \alpha) = \underbrace{\frac{\partial R_N}{\partial u} \frac{du}{d\alpha}}_{\mathbf{A}} + \frac{\partial R_N}{\partial \alpha} = 0 \quad \implies \quad \mathbf{A} \frac{du}{d\alpha} = -\frac{\partial R_N}{\partial \alpha}. \quad (14)$$

This is the implicit function theorem. It is the chain rule applied to a function that is being held at zero, not to a function whose value is being tracked, and the minus sign comes from that constraint: raising the coupling by itself would spoil the equations by $\partial_\alpha R_N$, and the metric must move by exactly the amount that cancels it. Where $\mathbf{A}$ is invertible — it is the same Jacobian the last Newton step already used — a single linear solve returns $du/d\alpha$, with no extrapolation and no step to refine.

What this buys is the missing ingredient of (12). Mass, entropy and each angular momentum are explicit functionals $\mathcal{F}[u; \alpha]$ of the discretised profile: the asymptotic fits of (6) and the horizon integral of (7). Their coupling derivatives follow by one more chain rule,

$$\frac{d\mathcal{F}}{d\alpha} = \frac{\partial \mathcal{F}}{\partial u} \cdot \frac{du}{d\alpha} + \frac{\partial \mathcal{F}}{\partial \alpha},$$

in which the second term is present only for the entropy, whose integrand (7) contains α explicitly as well as through the horizon geometry. Inserting $dM/d\alpha$, $dS/d\alpha$ and $dJ/d\alpha$ into (12) returns Ψ at that solution. Omitting the explicit ∂_α of the entropy is the easiest way to produce a smooth but wrong answer, so it is coded and tested separately.

The production code assembles $\mathbf{A}$ by the chain rule from local derivatives of the radial equations and horizon relations, and evaluates $\partial R_N/\partial \alpha$ the same way; appendix A gives the implementation and its verification. Equation (14) is an analytic differentiation identity for the discrete system, but its numerical solution still carries discretisation, solver and rounding errors, and the extracted observables carry asymptotic fitting errors as well, so what it delivers is an implicit numerical response and not an exact determination of Ψ .

The response is taken about solutions at finite α . It does not truncate the family at first order about $\alpha = 0$, which is what makes the measured dependence on the coupling nonlinear rather than perturbative.

As a check, we compare the response with central real differences of neighbouring solutions. At 4 points, using 5 step sizes related by halving, the discrepancy decreases by factors between 3.951 and 4.028, consistent with second-order convergence. Table 2 gives the locations, the first and last step sizes and the errors they leave rather than the ratios alone; the largest initial gap is 1.47×10^{-2} . Responses in both α and Ω_H also provide tangent predictors for the continuation towards extremality.

4 Results

The dataset contains 356 accepted solutions at 13 couplings from $\alpha = 0$ to $\alpha = 1/2$ in units $r_H = 1$. Every one of them is a solution of the nonlinear radial boundary-value problem of section 3.3, solved at that coupling: nothing is obtained by perturbing about $\alpha = 0$ or by rescaling the static

q	α	response Ψ	largest step		smallest step		ratios
			$\Delta\alpha$	error	$\Delta\alpha$	error	
0.3	0.02	-5.4014746	0.008	5.26×10^{-4}	0.0005	2.07×10^{-6}	3.951–4.014
0.6	0.02	-1.6298178	0.008	6.22×10^{-3}	0.0005	2.41×10^{-5}	4.001–4.017
0.65	0.02	-0.5008561	0.008	1.47×10^{-2}	0.0005	5.74×10^{-5}	4.001–4.010
0.65	0.06	-1.0520187	0.008	1.76×10^{-3}	0.0005	6.81×10^{-6}	4.002–4.028

Table 2. The implicit response against real central differences of neighbouring solutions. The error columns are the absolute difference between the differenced value and the response value quoted in the third column, at the two ends of a halving ladder of 5 steps; the last column gives the range of successive error ratios along that ladder. Four is the value expected of a second-order central difference. The response itself has no step size, so it has no entry in either error column.

solution. The spin coverage is not uniform across the 13 couplings, and the difference matters for what each figure can show. At 6 of them, namely $\alpha = 0, 0.02, 0.05, 0.1, 0.2, 0.5$, the family was walked over the whole spin range, from q near zero up to the near-extremal end. At the remaining 7 the walk starts at $q = 0.6$ and runs to the near-extremal end, which is the region the zero-temperature extrapolation of section 4.4 uses. Consequently table 4 has 13 rows while figure 2 and table 3, which need the low-spin end, have 6.

Figure 1 shows the four radial amplitudes themselves at all 13 couplings, at the one spin $q = 0.6$ that every walk passes through, so that the comparison isolates the coupling. These are independent re-solves along a different continuation route from the one used for the dataset — a spin ladder in the vacuum followed by a coupling ladder at fixed spin, rather than a response-seeded descent in the spin at fixed coupling — and they reproduce the mass, spin, entropy and temperature of the corresponding accepted states to 2.51×10^{-11} relative. The deformation the coupling produces is visible there and is not small. At $\alpha = 0$ the horizon three-sphere is squashed by $h(r_H)/r_H^2 = 1.5625$, which is the Myers–Perry value $1/(1 - q^2)$ and is reproduced by the solver rather than imposed. By $\alpha = 1/2$ the same ratio has fallen to 0.6129. It has crossed unity, so at the largest coupling the horizon is squashed in the opposite sense from its Myers–Perry counterpart at the same q .

The dataset reaches $x = 0.5001$ on the sampled family and $x = 0.4170$ on the extrapolated extremal branch. These are subsets of the allowed domain, whose bound $x < 1$ is reported in [2]. We follow the branch continuously connected to Myers–Perry. Continuation and regularity checks identify the branch sampled here; they do not establish global uniqueness or exhaust its domain.

4.1 An analytic example: the static limit

The static member is the Boulware–Deser solution [18]. In our conventions,

$$M = \frac{3\pi}{8}(r_H^2 + 2\alpha), \quad T = \frac{r_H}{2\pi(r_H^2 + 4\alpha)}, \quad S = \frac{\pi^2}{2}r_H^3 + 6\pi^2\alpha r_H. \quad (15)$$

This example also illustrates, in closed form, the constraints in (12). Differentiating (15) at fixed radius, the mass derivative $M_\alpha = 3\pi/4$ is positive, but the entropy is not held fixed: it changes by $S_\alpha = 6\pi^2 r_H$. There is no spin term, so subtracting the entropy contribution is all that (12) requires, and gives

$$\Psi_{\text{static}} = \frac{3\pi}{4} - T(6\pi^2 r_H) = \frac{3\pi}{4} \frac{4\bar{\alpha} - 3}{1 + 4\bar{\alpha}}, \quad \bar{\alpha} = \frac{\alpha}{r_H^2}. \quad (16)$$

In particular $\Psi_{\text{static}} \rightarrow -9\pi/4$ as $\alpha \rightarrow 0$. The mass derivative at fixed radius and the derivative at fixed entropy can have opposite signs; keeping the entropy fixed requires the radius to change.

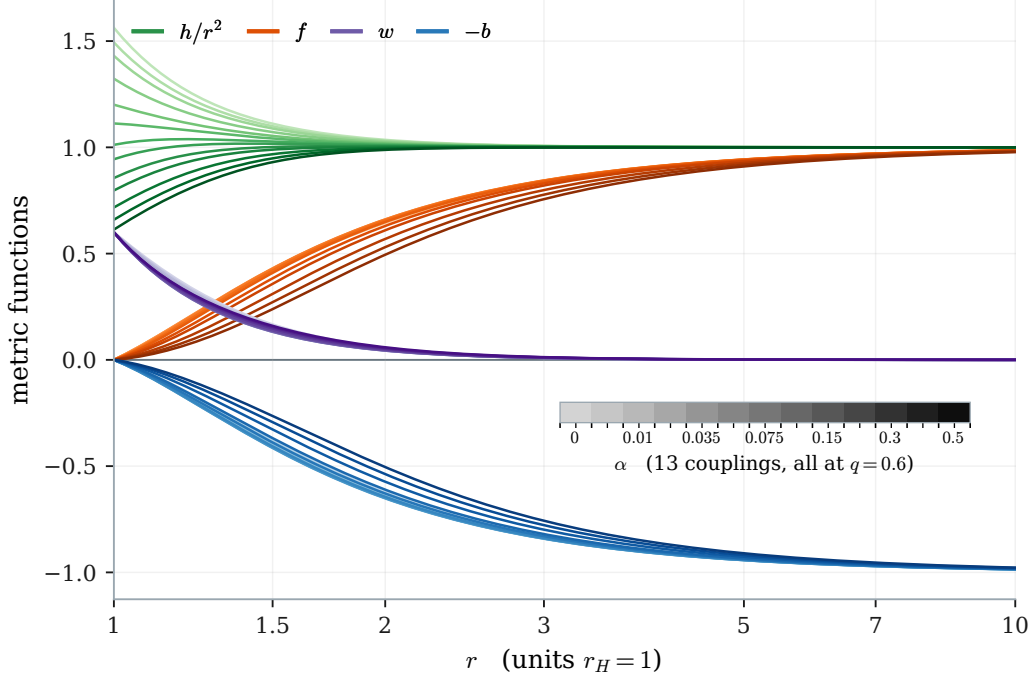


Figure 1. The four metric functions of (3) at all 13 couplings of table 4, all at the same spin $q = 0.6$ and in units $r_H = 1$. One colour per function, one shade per coupling, running from $\alpha = 0$ (lightest) to $\alpha = 1/2$ (darkest). Two conventions follow fig. 1 of [11] so that the two figures can be compared directly: h is shown as h/r^2 , which tends to unity at infinity whereas h itself grows like r^2 , and b as $-b$, because b and f otherwise overlap over most of the range. The boundary conditions are visible rather than imposed on the plot: $b = f = 0$ and $w = q$ at $r = r_H$, and $b, f, h/r^2 \rightarrow 1$, $w \rightarrow 0$ as $r \rightarrow \infty$.

Equation (16) satisfies (9) identically at zero spin.

Applying the numerical procedure at zero spin for 7 couplings reproduces the analytic mass, temperature, entropy and potential to worst relative deviations 3.20×10^{-11} , 1.08×10^{-13} , 1.00×10^{-16} and 2.60×10^{-9} , respectively. These are checks against an analytic solution at finite coupling. The potential changes sign at $\bar{\alpha} = 3/4$, or $x = 3/5$, on the static branch. That result alone does not fix the sign on all rotating solutions beyond the range sampled here.

4.2 The perturbative potential from the literature

The $\alpha = 0$ potential is known in closed form at every spin, and (17) below is neither a guess of ours nor a fit to our data: it is transcribed from a perturbative free energy in the literature, with the two steps of that transcription spelled out here.

The source is Wu and Lü [9], who compute the $\mathcal{O}(\alpha)$ correction to the thermodynamics of Myers–Perry black holes in a general quadratic-curvature theory, at fixed temperature and angular velocities, with the same $1/16\pi G$ action normalisation used in (2). Their result applies to the Gauss–Bonnet term because the Myers–Perry background is Ricci-flat: the R^2 and $R_{\mu\nu}R^{\mu\nu}$ terms of $\mathcal{L}_{\text{GB}}$ contribute nothing at this order, so at first order the coefficient of their $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ is our α .

The first step is thermodynamic. Define the Gibbs potential $\mathcal{G} = M - TS - \sum_i \Omega_i J_i$. The extended first law (8) gives $d\mathcal{G} = -S dT - \sum_i J_i d\Omega_i + \Psi d\alpha$, so the coefficient of α in $\mathcal{G}$ at fixed T and Ω_i is Ψ evaluated on the Einstein solution, which we call Ψ_0 . Equation (33) of the arXiv

version of [9], specialised to $D = 5$ with equal rotation parameters $a_1 = a_2 = a$, is then

$$\Psi_0 = \frac{\Delta\mathcal{G}}{\alpha} = -\frac{\pi}{4r_+^4}(a^4 - 14a^2r_+^2 + 9r_+^4),$$

with r_+ the Boyer–Lindquist horizon radius. The second step is the change of radial coordinate already recorded in (5), which turns this into a function of our spin label alone:

$$\Psi_0(q) = -\frac{\pi}{4}(u^2 - 14u + 9), \quad u = \frac{a^2}{r_+^2} = \frac{q^2}{1 - q^2}. \quad (17)$$

Equation (17) interpolates between the two ends of the vacuum family, and reproduces both independently known endpoint values although neither was used to obtain it. At $u = 0$ the solution is static, and (17) returns $-9\pi/4$, which is the $\alpha \rightarrow 0$ limit of the closed-form Boulware–Deser potential (16) obtained in section 4.1 from an entirely different calculation. At $u = 1$, which by (5) is $q = 1/\sqrt{2}$ and hence extremality, it returns π , the linear-order slope in (1) taken from [1]. Between them it changes sign once, at $q = 0.634936$. This is the analytic reference against which the numerical response is tested at zero coupling.

We evaluate the numerical response on the Myers–Perry background at $\alpha = 0$. It agrees with (17) to at least 8 significant digits at 9 of 11 sampled spins, with worst relative deviation 2.86×10^{-8} . No extrapolation in the coupling is needed, although the numerical response and charge extraction remain discretised.

4.3 The potential across the family

Figure 2 shows Ψ against t_H for the 6 couplings walked over the whole spin range. Every plotted point is one accepted rotating solution, and both of its coordinates are measured on that same solution: Ψ from its own linear response (14), and t_H from its own T and M through (11). Neither axis uses a static quantity. In particular t_H is not the static value $\sqrt{1 + 2\bar{\alpha}}/(1 + 4\bar{\alpha})$ held fixed along a curve; that value is the $q \rightarrow 0$ limit the curve runs towards, and the closed-form static potential (16) is plotted there as a cross so that the approach can be checked. The walks stop at a small nonzero spin, so the crosses lie just beyond the right-hand end of the curves rather than on them. The curves are therefore sequences of rotating solutions at fixed coupling, parametrised by spin, and not the static potential replotted.

On those sequences Ψ is negative near the static member and becomes positive towards extremality. Thus increasing α lowers the mass at fixed (S, J) in the former region and raises it in the latter. The location of the sign change depends on the coupling.

Every crossing in table 3 is bracketed by two accepted solutions of opposite sign, and both endpoints of the bracket are given there. The quoted position between them is an interpolant, so the digits it carries are not the width of the bracket and should not be read as one. What they can be calibrated against is the zero-coupling row, where the analytic zero (17) is known: the crossing is bracketed by solutions 0.02 apart in q and the interpolant lands 1.95×10^{-4} from the analytic value, so the interpolation is better than the sampling by two orders of magnitude, and it is that, rather than the bracket, that supports the digits printed. In the scaled spin the locus moves from $j = 0.9807$ to 0.4728. In temperature it moves from $t_H = 0.3245$ down to 0.1952 at $\alpha = 0.1$, then increases. A turning point in this coordinate does not by itself establish a common physical mechanism with the minimum of the extremal slope.

4.4 The extremal mass and its slope

The walks reach $\tau \leq 4.61 \times 10^{-3}$ at every coupling and 3.34×10^{-6} at the closest approach, in the scaled temperature τ of (10); the value at each coupling is the $\tau_{\min}$ column of table 4, which

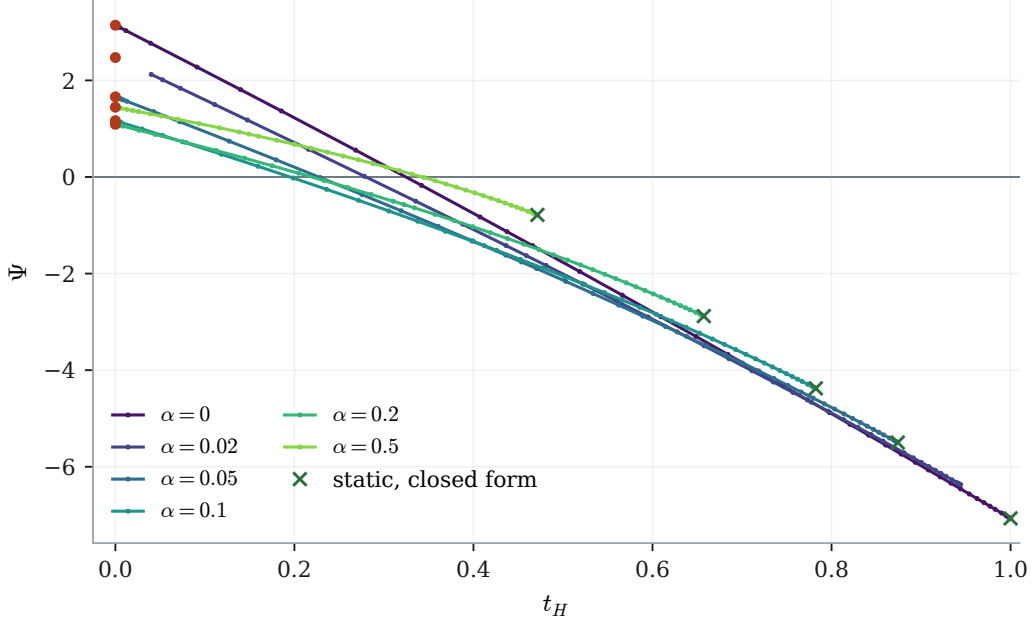


Figure 2. The measured potential along the 6 stationary families that were walked over the whole spin range; the other 7 couplings of table 4 do not reach the low-spin end and are not shown. Each point on a curve is one rotating numerical solution, with Ψ and t_H both evaluated on that solution. A curve ends on the left at the coldest state reached and on the right at the slowest rotating state reached; the crosses are the exact static values (16), plotted at the static temperature $t_H = \sqrt{1 + 2\bar{\alpha}}/(1 + 4\bar{\alpha})$, which is the $q \rightarrow 0$ limit each curve runs towards and not a point on it. Circles at $t_H = 0$ are the extrapolated extremal values of table 4, intercepts of fits rather than constructed solutions.

α	x	t_H	bracket width in t_H	j	q
0	0.0000	[0.63, 0.65]	0.63474	0.3245	0.9807
0.02	0.0247	[0.65, 0.67]	0.66943	0.2798	0.9433
0.05	0.0629	[0.71, 0.73]	0.71070	0.2272	0.9021
0.1	0.1240	[0.73, 0.75]	0.73172	0.1952	0.8554
0.2	0.2331	[0.67, 0.69]	0.67959	0.2192	0.7783
0.5	0.4829	[0.37, 0.39]	0.37294	0.3426	0.4728

Table 3. The measured sign-change locus, for the 6 couplings whose walks cover the low-spin end where Ψ is negative; at the other 7 no crossing is bracketed by accepted solutions and none is reported. Digits of the interpolant are supplied for reproducibility, not as an error estimate. At $\alpha = 0$ the crossing is $q = 0.63474$, compared with the analytic $q = 0.634936$.

is the scaled temperature of the coldest solution actually constructed there. The extrapolated branch is given in the same table. Central values are the $T = 0$ intercepts of cubic fits in the unscaled temperature T through the six coldest states at each coupling.

Every row of table 4 is an extrapolation from positive temperature, not a solution constructed at $T = 0$: no state in the dataset has zero temperature, and a row's temperature is zero by construction. What the table reports instead is how far the extrapolation had to reach, the column $\tau_{\min}$, which varies by orders of magnitude across the couplings; the rows with the largest $\tau_{\min}$ carry the largest sensitivity. The entries are therefore central extrapolants rather than measurements at extremality, and the extrapolation is the one step of this calculation that no check in section 5 repeats independently.

Figure 3 shows that step itself, which a table of intercepts hides. For each coupling it plots the

α	y	x	$\tau_{\min}$	μ	Ψ_{ext}	sensitivity	j	σ
0	0.00000	0.00000	1.12×10^{-5}	2.196888	3.14159	1.15×10^{-2}	1.00000	6.2832
0.005	0.00479	0.00510	2.44×10^{-3}	2.211579	2.99094	1.12×10^{-1}	0.99005	6.6686
0.01	0.00984	0.01042	1.21×10^{-3}	2.226279	2.82456	8.81×10^{-2}	0.98026	7.0685
0.02	0.02072	0.02165	4.61×10^{-3}	2.255058	2.47283	1.60×10^{-1}	0.96156	7.8933
0.035	0.03843	0.03946	3.76×10^{-3}	2.294438	2.00066	1.32×10^{-1}	0.93691	9.1213
0.05	0.05689	0.05758	4.49×10^{-4}	2.328017	1.65908	1.86×10^{-2}	0.91671	10.2781
0.075	0.08775	0.08711	4.38×10^{-4}	2.373482	1.32597	2.20×10^{-2}	0.89050	12.0267
0.1	0.11787	0.11521	3.84×10^{-4}	2.410729	1.16621	2.10×10^{-2}	0.86994	13.5922
0.15	0.17520	0.16686	2.57×10^{-4}	2.473993	1.07149	1.30×10^{-2}	0.83679	16.3436
0.2	0.22906	0.21316	1.44×10^{-4}	2.531936	1.09007	1.12×10^{-2}	0.80823	18.7577
0.3	0.32921	0.29309	1.06×10^{-4}	2.646582	1.20781	6.16×10^{-3}	0.75628	22.9735
0.4	0.42251	0.36002	5.06×10^{-5}	2.765180	1.33384	6.96×10^{-3}	0.70815	26.6807
0.5	0.51124	0.41702	3.34×10^{-6}	2.888535	1.44440	1.39×10^{-3}	0.66328	30.0608

Table 4. The extrapolated extremal branch; these rows are $T = 0$ intercepts of fits and not solutions, so the temperature of a row is zero by construction. What is reported instead is $\tau_{\min} = \min T J^{1/3}$, the scaled temperature of the coldest solution the walk at that coupling actually reached. The sensitivity refers to Ψ_{ext} and to no other column; it is defined in section 4.4 and constructed in appendix A. The first row is compared with $\mu = (3/2)\pi^{1/3}$ and $\Psi_{\text{ext}} = \pi$ in section 5.

six states the production fit uses, the cubic fitted through them, the intercepts returned by the other twenty fits of the sensitivity list, and the residuals the production fit leaves. Three things are visible there. The fit describes its own points to 1.77×10^{-6} or better at every coupling, which is small compared with every sensitivity in table 4; so the envelope in that table is not driven by a poor fit but by the length of the extrapolation. The distance the fit travels differs by more than an order of magnitude between couplings, and the four whose curves travel furthest are precisely the four carrying the largest sensitivities in table 4. And the competing intercepts cluster at the axis rather than scattering, so the spread the table reports is a spread between models that mostly agree, not a disagreement about the value.

What the figure cannot show is a bias shared by all twenty-one fits. We have not derived the leading near-extremal powers of T for this system, so the cubic is a convenience and not a model of the approach to extremality; the square-root and logarithmic probes in table 6 test for two specific departures from analyticity and not for the general case. One further stability test is reported in appendix A: refitting with each of the six states removed in turn moves the intercept by at most 1.09×10^{-5} , so no single cold or warm state carries the result.

The *sensitivity* column answers one question: by how much does that intercept move when the fit is changed? We refit the same states with polynomials of degree one, two and three, on the six coldest states and on up to the twelve coldest, in the coordinate T and in the coordinate τ , and with two non-polynomial models; the quoted half-width is the largest displacement of the intercept over that whole list. It measures sensitivity to a stated set of modelling choices: it carries no confidence level, is not a standard error, and cannot detect an error common to every fit in the list, such as a systematic error in the underlying charges. Every band and spread quoted below is of this kind. The full list is in appendix A; the contribution of the two non-polynomial models, which dominates the envelope at most couplings, is tabulated there in table 6.

The main qualitative result is visible in figure 5: the slope Ψ_{ext} is positive throughout the sampled range, falls below its perturbative value, and subsequently increases. The lowest sampled central value is 1.07149 at $y = 0.1752$, with fit sensitivity 1.30×10^{-2} . Its neighbouring samples lie at $y = 0.118$ and 0.229 . The last sampled value is 1.44440 at $y = 0.5112$. The suppression at the lowest sampled point relative to π is about a factor 2.93. We do not assign a precise location or confidence interval to a continuous minimum between the sampled couplings.

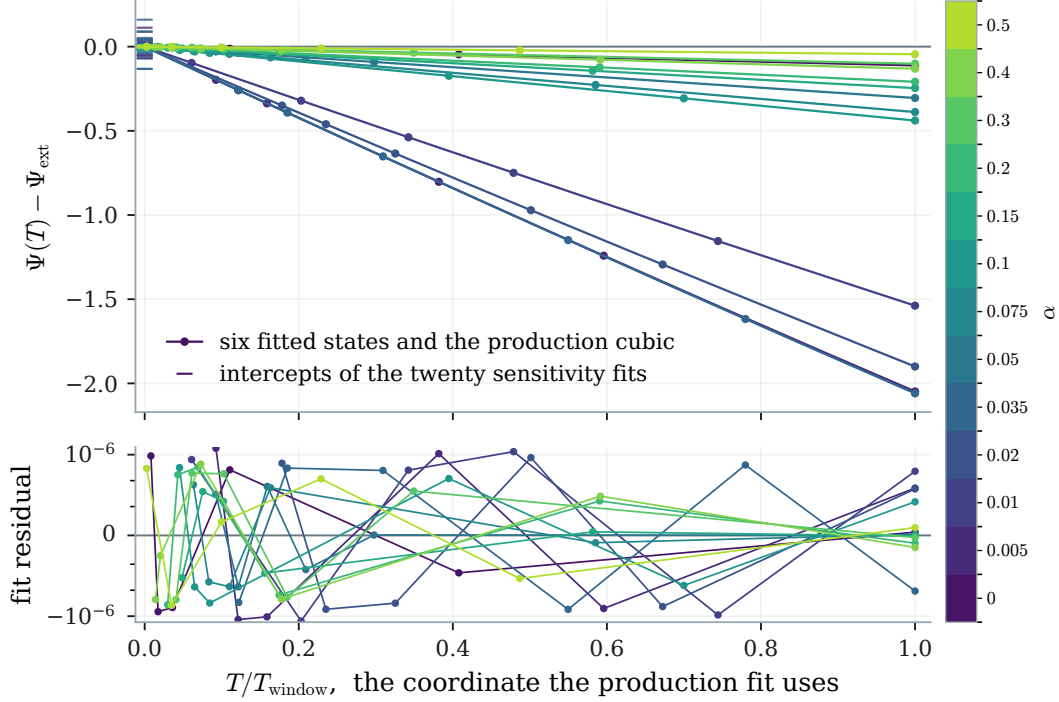


Figure 3. The zero-temperature extrapolation, which produces every row of table 4. Filled circles are the 6 coldest accepted solutions at each coupling and the line through them is the production cubic; the abscissa is the temperature divided by the temperature of the warmest fitted state, the coordinate the fit itself uses, so that windows spanning four decades of T can be compared, with the physical temperatures given by the $\tau_{\min}$ column of table 4. The central value is subtracted, so every curve reaches the origin by construction and the figure shows how far it travelled to get there. Short horizontal ticks on the left axis are the intercepts of the other twenty fits in the sensitivity list; the envelope of those ticks is the sensitivity column of table 4, and the lower panel gives the residuals of the production fit on a symmetric logarithmic scale.

The central extrapolated M_{ext} remains increasing at fixed J across the sampled range: the μ column of table 4 is monotone, and its increments exceed the corresponding spreads in table 7 by two orders of magnitude or more. The accumulated mass change is the integral of the slope:

$$M_{\text{ext}}(J, \alpha) - M_{\text{ext}}(J, 0) = J^{2/3} \int_0^{\alpha/J^{2/3}} \Psi_{\text{ext}}(y') dy'.$$

Thus the suppression of the local slope is not a suppression factor for the total mass shift.

Figure 4 shows the mass that this integral accumulates. Across the sampled range μ rises by 31%, while the first-order expression (1) runs above it and overshoots by a factor 1.32 at the largest coupling. The fall of the slope below π is a fall in the rate: the extremal mass grows more slowly than linear order predicts, not more slowly than at $\alpha = 0$. The right-hand panel places the same branch in the (x, j) plane of [2], whose reported domain of existence is bounded by $x < 1$ and $j \leq 1$; our branch stays inside both. Scale invariance ties the two panels, since $M_{\text{ext}} = J^{2/3} \mu(y)$ gives

$$j_{\text{ext}}(y) = \left[\frac{\mu(0)}{\mu(y)} \right]^{3/2}, \quad \mu(0) = \frac{3}{2} \pi^{1/3}, \quad (18)$$

which our separately extrapolated j and μ satisfy to 3.11×10^{-11} . The decrease of j along the

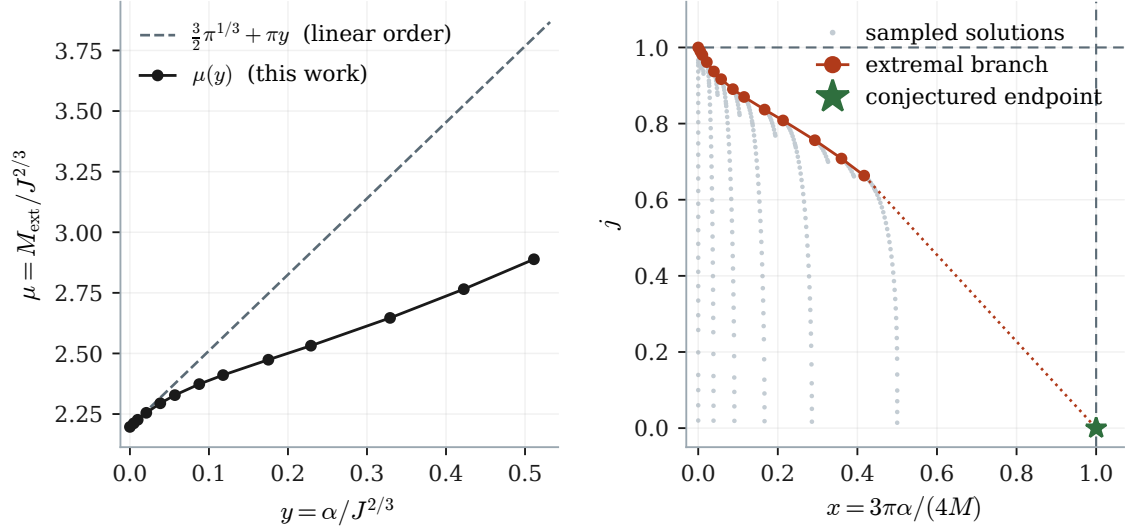


Figure 4. Left: the extremal mass in scale-invariant form, $\mu = M_{\text{ext}}/J^{2/3}$ against $y = \alpha/J^{2/3}$, with the first-order line (1) dashed. Right: the same branch in the (x, j) plane of [2], with every accepted non-extremal solution in grey, the bounds $x < 1$ and $j \leq 1$ reported there dashed, and the endpoint conjectured there as a star; the dotted segment joins our last sampled point to that endpoint and is not computed. Every plotted extremal point is a $T = 0$ intercept of a fit and not a constructed solution.

branch is therefore the increase of μ re-expressed, and not an independent measurement.

We compare the potential with a second quantity formed from the extrapolated masses and spins. Scale invariance implies

$$\int_{y_1}^{y_2} \Psi_{\text{ext}}(y) dy = \mu(y_2) - \mu(y_1). \quad (19)$$

The right side does not use the entropy formula, so this provides a useful consistency test. It is not an independent determination: both sides share masses, spins, the solver and the temperature-based selection and extrapolation of the near-extremal states.

We integrate a local cubic interpolant through the four nearest potential samples. The largest interval discrepancy in (19) is 1.64×10^{-4} , or 1.96×10^{-3} relative to the corresponding change in μ . Two different cumulative figures have to be distinguished. The discrepancy accumulated across the whole range, that is its value at the last interval $y = 0.5112$, is 3.81×10^{-4} on a total change 0.6916; the largest *running* value, reached before the last interval and then partly cancelled, is 4.38×10^{-4} . Using trapezoids instead gives a largest interval discrepancy 8.76×10^{-4} . The comparison includes quadrature error and correlated errors in the observables. Propagating the recorded extrapolation spreads of both routes, the discrepancy is within the combined spread at 12 of the 12 intervals, the worst case reaching 0.90 of it; since the two routes share those spreads this is a consistency statement about the quadrature and not an independent confirmation. Agreement of the changes in μ does not bound a common offset in the masses, and we do not use it as an absolute mass-accuracy estimate or a calibration of the sensitivity bars.

A one-sided quadratic fit to the lowest-coupling mass samples gives $\mu'(0) = 3.144521$, differing from π by 2.93×10^{-3} . Varying the number of samples gives a spread 2.09×10^{-2} . This derivative uses the mass curve rather than the entropy formula, but still relies on the thermodynamic extraction and extremal extrapolation.

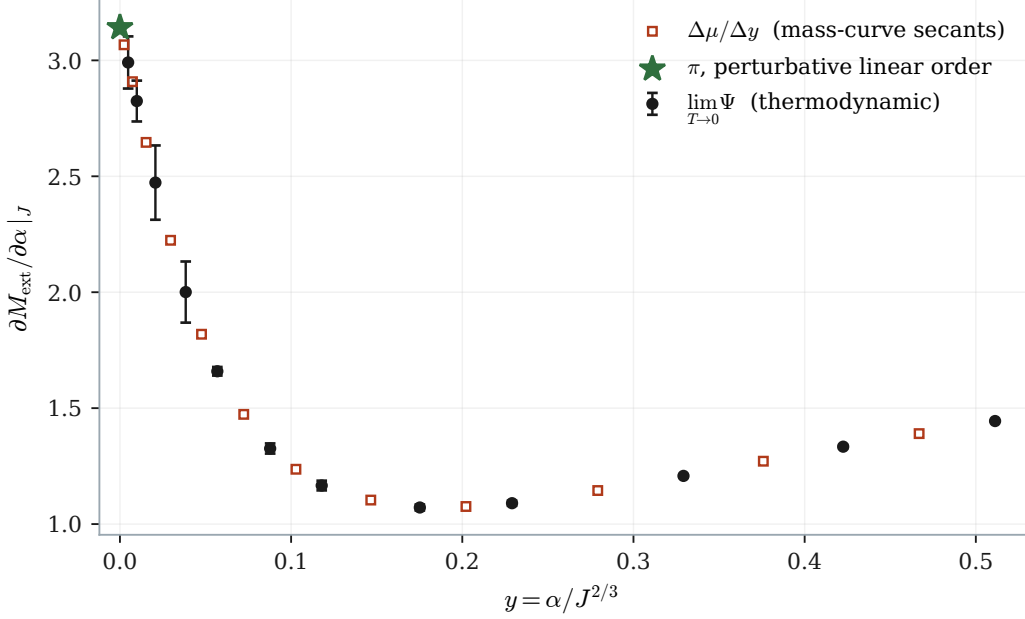


Figure 5. The extremal mass slope. Circles show the extrapolated potential with its fit-sensitivity envelope; open squares show secants of $\mu(y)$ at interval midpoints. A secant is an interval average rather than a pointwise derivative, and the two estimates share numerical inputs. The star is the perturbative value π of [1], which is not imposed in the response calculation.

5 Checks against known results

The analytic static and perturbative checks were described above. Here we compare the extremal mass and entropy with results from the literature and distinguish those comparisons from internal consistency tests.

Einstein and perturbative extremal limits. At $y = 0$ we obtain

$$\mu = 2.196887831, \quad \frac{3}{2}\pi^{1/3} = 2.196887831, \quad \Psi_{\text{ext}} = 3.141592534.$$

The absolute deviations from their respective analytic values are 4.56×10^{-11} and 1.19×10^{-7} , respectively. These numbers result from the same continuation and extrapolation used at finite coupling; the limiting constants are not fitted constraints. The agreement at $y = 0$ does not by itself establish the error at finite y .

The potential from the Smarr relation. The response calculation of section 3.3 is the one step of this work with no analytic counterpart at finite coupling and finite spin, so it is worth checking against something that does not use it. The Smarr relation (9) is such a thing. Read as an equation for Ψ rather than as a residual, it gives

$$\Psi_{\text{Smarr}} = \frac{2M - 3TS - 6\Omega_H J}{2\alpha}, \quad (20)$$

which involves only the charges of a single solution: no Jacobian, no linear solve, no parameter derivative and no second solution. It is therefore a genuine test of the implicit-response machinery — of (14), of the chain-rule assembly, of the complex-step parameter derivative and of the explicit ∂_α of the entropy — and not a repetition of it. What it does share with the response is the set of solutions and the asymptotic charge extraction, so a bias in either would move both.

α	$\Psi_{\text{ext, response}}$	$\Psi_{\text{ext, Smarr}}$	relative difference
0.005	2.990943	2.990939	1.59×10^{-6}
0.01	2.824558	2.824558	8.73×10^{-9}
0.02	2.472827	2.472827	1.64×10^{-8}
0.035	2.000660	2.000660	1.02×10^{-8}
0.05	1.659080	1.659079	3.75×10^{-7}
0.075	1.325970	1.325969	1.86×10^{-7}
0.1	1.166214	1.166214	7.16×10^{-8}
0.15	1.071492	1.071493	2.05×10^{-7}
0.2	1.090070	1.090071	8.55×10^{-7}
0.3	1.207813	1.207814	1.21×10^{-6}
0.4	1.333842	1.333843	7.56×10^{-7}
0.5	1.444398	1.444398	2.63×10^{-8}

Table 5. The extremal potential by two routes. The first column repeats table 4; the second extrapolates (20), evaluated state by state from the charges alone, to $T = 0$ with the same cubic through the same 6 states. The two share the solutions and the charge extraction but not the response solve. There is no $\alpha = 0$ row because (20) divides by the coupling.

Over the 313 accepted states at $\alpha \neq 0$, (20) reproduces the response value to a worst relative difference 1.45×10^{-6} , worst absolute difference 3.30×10^{-6} . Extrapolating Ψ_{Smarr} to $T = 0$ by the same cubic used for the production values gives the comparison in table 5: the two routes to Ψ_{ext} agree to 1.59×10^{-6} relative at all 12 finite couplings, well below every fit sensitivity in table 4. The division by α in (20) is ill-conditioned as $\alpha \rightarrow 0$, which is why no row appears at $\alpha = 0$ and why this route cannot replace the response calculation; at $\alpha = 0$ the corresponding test is the perturbative potential of section 4.2.

The potential as an on-shell action. At fixed temperature and angular velocities the Gibbs potential of a stationary black hole is its on-shell action per unit time, and the only *explicit* dependence of that action on α is the Gauss–Bonnet term of (2). Differentiating at fixed (T, Ω_i) and discarding the implicit variation with the field equations gives

$$\Psi = -\frac{1}{16\pi} \int_{\Sigma} d^4x \sqrt{-g} \mathcal{L}_{\text{GB}}, \quad (21)$$

with Σ a stationary spatial slice outside the horizon; the integral converges because $\mathcal{L}_{\text{GB}}$ falls off like r^{-8} on an asymptotically flat solution while $\sqrt{-g}$ grows like r^3 . The boundary terms of the action carry α too, and that their net contribution to (21) vanishes for this family is part of what the test below confirms. This route uses no Jacobian, no linear solve, no parameter derivative and — unlike (20) — no asymptotic charge extraction. It reads Ψ off the curvature of the solution itself, which is why it tests a different part of the chain than either of the two checks above.

Evaluating the integrand with the same independent curvature evaluator that gates acceptance, on the 43 archived spectral profiles, reproduces the response value to a worst relative difference 4.67×10^{-7} and a worst absolute difference 4.01×10^{-8} . The quadrature is described in appendix A. Those states reach $\alpha = 0.75$ but only $q = 0.4$: the near-extremal states that carry the zero-temperature extrapolation are not among them, so this check covers the potential across the coupling range and not at the extremal end.

Extremal entropy. The near-horizon entropy function determines S_{ext}/J as a function of y without integrating the radial equations. We rederived it rather than transcribing it, as summarised in appendix B, and checked the rederivation against the equations of [11]; the curve in

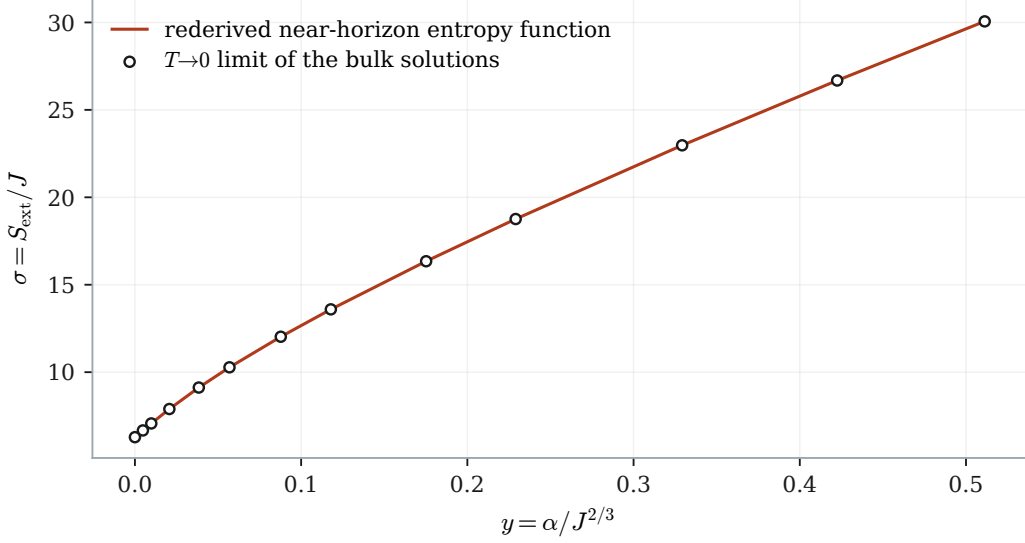


Figure 6. Extremal entropy per angular momentum. The line joins values from our rederived near-horizon entropy function, checked against the equations of [11] to the precision quoted in the text; the points are extrapolations of bulk solutions to zero temperature.

figure 6 is that rederived solution and not a quotation. Solving its algebraic equations numerically at finite coupling gives the comparison shown there; the worst relative difference from the extrapolated bulk entropy is 4.06×10^{-6} . This validates the extremal entropy at finite coupling, but does not provide a near-horizon measurement of the asymptotic mass.

In the variables of [11], our near-horizon solutions satisfy its eqs. (4.11) and (4.12) to 9.17×10^{-14} and 6.66×10^{-16} , respectively. This comparison checks the coupling conversion $\alpha_{\text{ref.}} = 4\alpha$ of table 1. One comparison with the same reference does not close; it is discussed in section 6.

Internal thermodynamic checks. At 10 selected states we evaluate the first law in the spin direction at fixed coupling:

$$\partial_q M - T \partial_q S - 2\Omega_H \partial_q J = 0.$$

Its worst residual is 1.35×10^{-8} relative to the largest term. We also evaluate $-T \partial_\alpha S|_{M,J}$ using coupling and spin responses and the scale direction. It agrees with Ψ to 4.84×10^{-8} relative. This is an internal check constructed from the same observables, not an independent test of the first law. No degeneracy is implied by the static value $J = 0$: generically J vanishes linearly with spin and its spin derivative there is nonzero.

6 Discussion

The observable calculated here is the response of mass to the Gauss–Bonnet coupling at fixed entropy and angular momenta, and its zero-temperature limit is the response of the extremal mass at fixed angular momenta, provided the branch has a regular thermodynamic limit. Stated at the precision the evidence supports: on the equal-spin branch continuously connected to five-dimensional Myers–Perry, the positive-temperature solutions and their zero-temperature extrapolations give a $\partial M_{\text{ext}}/\partial \alpha|_J$ that is positive over the sampled range, falling from the perturbative value π to a broad low region near 1.07 and rising again to 1.44. The positive initial slope was already known; the fall and the subsequent rise are the finite-coupling behaviour investigated here. Both moves exceed the fit sensitivities of the rows involved by more than an order of

magnitude. What those sensitivities do not resolve is where the turning point lies between the sampled couplings, which is why we quote a lowest sampled value rather than a minimum.

Slope and accumulated mass change have to be kept apart. At fixed J the central extrapolated extremal mass increases throughout the range measured; only its rate of increase turns over, reaching its lowest sampled value near $\alpha = 0.15$ in units $r_H = 1$, corresponding to $x = 0.1669$. By $x = 0.0871$ the slope is already 1.3260, below half the perturbative value. Outside this interval the data establish neither the sign nor the shape.

6.1 Interpretation

What a positive slope says. Two restatements make the sign concrete. The first concerns the spin bound: by (18) a positive μ' is the same statement as a decreasing j_{ext} , so the largest spin a black hole of given mass can carry falls as the coupling grows. The bound $j \leq 1$ reported in [2] is the integrated form of the sign measured here pointwise. The second concerns the horizon: setting $T = 0$ in the Smarr relation (9) gives

$$\alpha \Psi_{\text{ext}} = M_{\text{ext}} - 3\Omega_H J, \quad (22)$$

so $\Psi_{\text{ext}} > 0$ is exactly $M_{\text{ext}} > 3\Omega_H J$, with the Gauss–Bonnet term making up the difference between the extremal mass and the rotational term of the Smarr relation. Both sides of (22) vanish at $\alpha = 0$, where extremal Myers–Perry satisfies $M_{\text{ext}} = 3\Omega_H J$ exactly; the first-order result $\Psi_{\text{ext}} \rightarrow \pi$ of [1] is then the statement that the excess $M_{\text{ext}} - 3\Omega_H J$ opens linearly in the coupling, and our measurement is that it keeps opening, at a rate that falls and then recovers.

The sign of the potential. Equation (21) says what the sign of Ψ is in terms of the geometry: a positive Ψ means the Gauss–Bonnet density integrated over the exterior of a stationary slice is negative. The potential is negative near the static, slowly rotating members of each family and positive towards extremality, so the locus of table 3 is where that integrated density changes sign, and $x = 3/5$ on the static branch of section 4.1 is the same crossing at zero spin. The first law also gives $\partial S / \partial \alpha|_{M,J} = -\Psi/T$, so the same sign says that a negative integrated Gauss–Bonnet density is a decrease of entropy at fixed mass and angular momenta. This is consistent with the entropy per spin along the *extremal* branch growing by a factor 4.8 over the sampled range, because extremal states at fixed J have different masses as α changes.

Where the slope is smallest. Dividing (22) by $J^{2/3}$, writing $\omega = \Omega_H J^{1/3}$ for the scale-invariant horizon angular velocity and using $\Psi_{\text{ext}} = \mu'$ gives

$$3\omega(y) = \mu(y) - y \mu'(y), \quad \text{hence} \quad 3\omega'(y) = -y \mu''(y). \quad (23)$$

For $y > 0$ the extremal branch therefore has a stationary ω exactly where the slope is stationary, and a minimum of μ' is a *maximum* of ω : the inflection of the extremal mass is the extremal horizon with the largest angular velocity in units of $J^{-1/3}$. That coupling is not otherwise distinguished, and in particular is not a boundary of the domain of existence; the sign-change locus of table 3 turns over at the neighbouring sampled coupling $\alpha = 0.1$, and the grid does not resolve whether the two coincide. Extrapolating ω on its own from the same walks with the production model gives a maximum $\omega = 0.762$ at $y = 0.175$, the same sampled coupling at which Ψ_{ext} is smallest, and satisfies (23) to 5.74×10^{-6} along the branch; at $y = 0$ it returns 0.732296 against the extremal Myers–Perry value $\pi^{1/3}/2$, to 1.16×10^{-13} .

Why the slope has to rise again. The mass of a static Gauss–Bonnet black hole is bounded below by that of the nakedly singular configuration, $M > 3\pi\alpha/4$, and [2] finds the same bound on

the rotating solutions; in scale-invariant form it is $\mu(y) > 3\pi y/4$. Suppose the slope stayed below some $c < 3\pi/4$ for all y . Then $\mu(y) \leq \mu(0) + cy$, which violates the bound once $y > \mu(0)/(3\pi/4 - c)$. The bound therefore forces μ' to return to at least $3\pi/4 = 2.356$, a value above every slope we measure, so the rise from 1.07 to 1.44 over the upper part of our range is required and not accidental. If in addition the extremal branch terminates as conjectured in [2], at $M = 3\pi\alpha/4$ with $J \rightarrow 0$, then $\mu(y)/y \rightarrow 3\pi/4$ and the average slope tends to 2.356, as does μ' itself wherever it converges. Our sampled range stops well short of that: the largest x reached on the extremal branch is 0.4170, against the conjectured endpoint at $x = 1$. The measured rise is consistent with this picture and does not establish it.

A constraint distinction at low spin. Holding the coordinate radius r_H fixed does not hold the entropy fixed. At $\alpha = 1/2$ the measured mass initially decreases slightly with spin before increasing, which the first law permits when the negative entropy contribution exceeds the rotational one; the parity of $S(q)$ and $J(q)$ does not determine the sign of $\partial_q M$. We therefore do not read a mass below the static value at the same r_H as an error, nor use that static value as a lower bound on the rotating family. The bounds $x < 1$ and $j \leq 1$ of [2] are different statements: our largest scaled spin is 1.000000, and the extremal value falls to 0.66328 at the largest sampled coupling.

6.2 Limitations

The leading limitation is the extremal extrapolation, and the displayed bands measure sensitivity to specified fitting choices rather than numerical error. Radial resolution is not the leading term: the complete ascending sequence and its $T = 0$ extrapolation were repeated at two coarser base resolutions at every one of the 13 couplings, including the three samples at $y = 0.118, 0.1752$ and 0.229 that bracket the turn in the slope, and the extremal intercept moves by at most 6.89×10^{-5} , below the fit sensitivity at the same coupling in 26 of the 26 sequences. Appendix A reports that study observable by observable and names the three reasons it bounds the resolution effect instead of proving convergence; small field-equation residuals do not by themselves bound the error in M , J , S or Ψ , and what bounds it here is the recomputation. The coupling coverage is incomplete and the spin coverage uneven, the low-spin end and hence the sign-change locus being mapped at only 6 of the 13 couplings, and the whole calculation is restricted to equal spins. We do not show that the branch followed here is unique, stable, or the lowest in mass among all five-dimensional solutions at the same angular momenta, and we claim no global minimum mass anywhere. None of the checks closes the gap that matters most: the mass-curve comparison shares systematic errors with the potential, the near-horizon route determines entropy rather than mass because the near-horizon geometry discards the asymptotic normalisation, and the Smarr and on-shell routes remove the response solve from the chain but not the solutions themselves. Finally, the calculation treats Einstein–Gauss–Bonnet as a classical theory with the action (2): second-order equations do not by themselves establish existence, uniqueness, stability or well-posedness at every coupling, and in a gravitational effective field theory terms beyond Gauss–Bonnet contribute at the same order as the higher powers of α kept here, so the non-monotonic slope is not a universal prediction of a derivative expansion. Causality restrictions on higher-curvature graviton interactions are a further reason to separate this classical truncation from an ultraviolet completion [19].

One comparison with [11] does not close. A separate reconstruction of the non-extremal family gave an outer ergosurface radius 1.102101 against the 1.104 quoted there, at $\alpha = 1/2$ in our convention. The discrepancy is unresolved and we do not offer it as evidence of an erratum. It tests the reconstruction of the non-extremal metric functions away from the horizon; it does not test the asymptotic charge extraction, the response solve or the extrapolation, none of which uses the ergosurface radius, and the near-horizon comparison of section 5, which does test the

coupling conversion, agrees to 9.17×10^{-14} .

Three things would sharpen the result. A direct construction of extremal solutions carrying their own response measurement, along the lines of [2], would remove the extrapolation; a derivation of the leading near-extremal powers of T would do more still, since it would replace the fitted cubic with a model; and a second-order perturbative calculation would test the initial departure from $\Psi_{\text{ext}} = \pi$ independently.

A Numerical construction and sensitivity analysis

Radial discretisation. We set $r_H = 1$, introduce $z = 1/r$ and collocate in $\chi = 1 - z$. The horizon is at $\chi = 0$ and infinity at $\chi = 1$. The unknown amplitudes factor out the boundary behaviour:

$$b = (1 - z^2)B, \quad f = (1 - z^2)F, \quad h = \frac{1 + z^4 H}{z^2}, \quad w = z^4 W.$$

They are represented at Chebyshev–Lobatto nodes. At infinity, $\chi = 1$, we impose $B = F = 1$ and $H' = W' = 0$, primes denoting χ -derivatives. At the horizon, $\chi = 0$, where the solved interior equations are singular, we impose three regularity relations together with $W(\chi = 0) = \Omega_H$, which is what fixes the spin of the solution. The equations and horizon relations are generated symbolically. Damped Newton iteration solves the collocation system. Production walks use $N = 64$, with $N = 96$ available when acceptance requires refinement; $N = 48$ is used for selected comparison walks.

Jacobian and response. The production Newton and response solves use the same chain-rule assembly of $\partial R_N / \partial u$. Local partial derivatives are computed by complex step: seven vectorised partials of the interior equations and nine of the horizon relations. The coupling and angular-velocity derivatives are computed by complex step in the respective parameter, with $\epsilon = 10^{-20}$. For a real analytic function, $\text{Im } f(p + i\epsilon)/\epsilon = f'(p) + \mathcal{O}(\epsilon^2)$, so a small imaginary step gives the derivative without subtracting nearly equal real numbers and without a step size to refine. The rational residual is evaluated using complex arithmetic without non-analytic operations on the perturbed variables. The assembled Jacobian is checked entrywise against a dense complex-step perturbation of every nodal unknown, including profiles that do not solve the equations. This checks the chain-rule assembly; both implementations still use complex-step differentiation.

Continuation. Starting from Myers–Perry, we continue in coupling and spin. The tangent predictor $u + \Delta\alpha \partial_\alpha u + \Delta q \partial_q u$ provides the initial guess for each nonlinear solve. In the tested near-extremal sequences it reaches states at which reusing the previous solution failed to converge. The predictor is only an initial guess: every retained state must satisfy the same acceptance checks.

Acceptance and conditioning. An evaluator separate from the generated radial equations constructs curvature tensors from the metric and checks $G_{\mu\nu} + \alpha H_{\mu\nu}$ at 51 off-grid points. We divide the maximum absolute residual over these points by the maximum reference scale over the same points. The reference scale is the largest of the Einstein and Gauss–Bonnet tensor contributions and the square root of the absolute Kretschmann invariant. The relative threshold is 10^{-8} ; the absolute residual is recorded as well. The largest relative residual over the accepted states is 9.76×10^{-9} , and the largest relative Smarr residual is 1.64×10^{-7} . These residuals diagnose equation closure; they are not bounds on the error of an extracted observable.

The largest condition number in the selected 10 checks is 5.4×10^7 . It is $\|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2$ for the assembled $\mathbf{A} = \partial R_N / \partial u$ in the collocation variables (B, F, H, W) at the Chebyshev–Lobatto

nodes, with no row equilibration and no rescaling of the unknowns; the quantity is therefore tied to that choice of variables and is not comparable with a condition number reported by a solver using different ones. Multiplying a condition number by machine precision estimates possible error amplification, not the number of correct digits in Ψ . These selected checks do not bound conditioning at all near-extremal states or at every resolution.

Asymptotic charge extraction. The coefficients in (6) are obtained by fitting compactified amplitudes in z^2 . The three windows and degrees are $z \in [0.03, 0.12]$ at degree 4, $z \in [0.04, 0.16]$ at degree 4, $z \in [0.03, 0.12]$ at degree 3. The first is the production choice; the other two measure sensitivity to that choice. We also compare masses extracted from the b and f asymptotics. The largest recorded relative diagnostic is 1.31×10^{-8} . All these extractions can share a bias, so a small spread does not certify their absolute accuracy. At $\alpha = 0$, comparison with the analytic $M = 3\pi r_H^2/[8(1 - q^2)]$ gives a worst relative deviation 3.27×10^{-13} through $q = 0.7$. The static finite-coupling comparison is in section 4.1. We have no corresponding analytic rotating mass at finite coupling.

The on-shell integral. Equation (21) is evaluated on the archived spectral profiles, using the same independent curvature evaluator as the acceptance gate. On the ansatz (3) the integrand factorises as $\sqrt{-g} \mathcal{L}_{\text{GB}} = \sin \theta \cos \theta D(r)$, which the run checks at four polar angles rather than assuming; the angular integral is then $2\pi^2$ and $\Psi = -(\pi/8) \int_{r_H}^{\infty} D dr$. The radial integral uses Gauss–Legendre quadrature in the compactified coordinate $\chi = 1 - 1/r$ with 80 nodes. It cannot be refined indefinitely: $\mathcal{L}_{\text{GB}}$ is a difference of curvature squares that cancels to $\mathcal{O}(r^{-8})$, so nodes pushed towards $\chi = 1$ lose the density to floating-point cancellation while $dr/d\chi$ amplifies what survives. The node count is therefore a recorded parameter and the convergence ladder is stored with the result.

Central extrapolation. At each coupling, states are sorted by the physical temperature T in units $r_H = 1$. The central estimates of M, J, S, Ψ are the intercepts of cubic fits through the six coldest states. The original degree checks use linear, quadratic and cubic fits through four, five and six states, respectively. The invariants y, μ, σ are then formed from the extrapolated charges. The use of unscaled T here is distinct from the use of $\tau = TJ^{1/3}$ to report the distance to extremality.

Fit-sensitivity envelopes. For the potential we recompute fits on the six coldest states and on up to twelve coldest states, using both T and τ . For each coordinate and window we use polynomials of degree one, two and three and the forms

$$c_0 + c_1 v + c_{1/2} \sqrt{v}, \quad c_0 + c_1 v + c_2 v^2 + c_{\log} v^2 \log v,$$

where v is the chosen coordinate divided by its largest value in the fitting window. This rescaling improves conditioning. The half-width displayed in table 4 and figure 5 is the largest deviation of any of these intercepts from the production central estimate, or the original degree spread if larger. The central values are unchanged by this reanalysis.

Table 6 shows the largest changes from the square-root and logarithmic models across the two coordinates and windows. These terms are sensitivity probes, not a derivation of the near-extremal expansion. Small changes for a given probe do not exclude other non-analytic behaviour. The envelope also includes the polynomial window and coordinate changes.

Removing a fitted state. The envelope above varies the model, the window and the coordinate but always fits the same states. A separate test varies the states: we refit the production cubic 6

α	$T_{\min}$	$T_{\max}/T_{\min}$	square-root shift	logarithmic shift
0	1.08×10^{-5}	3919	1.07×10^{-2}	1.11×10^{-4}
0.005	2.40×10^{-3}	24	1.12×10^{-1}	2.88×10^{-3}
0.01	1.21×10^{-3}	49	8.81×10^{-2}	2.37×10^{-3}
0.02	4.73×10^{-3}	13	1.60×10^{-1}	8.97×10^{-3}
0.035	3.96×10^{-3}	15	1.32×10^{-1}	7.61×10^{-3}
0.05	4.80×10^{-4}	86	1.86×10^{-2}	3.59×10^{-4}
0.075	4.75×10^{-4}	86	2.20×10^{-2}	4.31×10^{-4}
0.1	4.17×10^{-4}	94	2.10×10^{-2}	4.17×10^{-4}
0.15	2.78×10^{-4}	107	1.30×10^{-2}	2.01×10^{-4}
0.2	1.55×10^{-4}	183	1.12×10^{-2}	1.85×10^{-4}
0.3	1.11×10^{-4}	173	6.16×10^{-3}	8.20×10^{-5}
0.4	5.20×10^{-5}	392	6.96×10^{-3}	1.29×10^{-4}
0.5	3.38×10^{-6}	2900	1.38×10^{-3}	1.24×10^{-5}

Table 6. Sensitivity of the potential intercept to non-polynomial fit models. $T_{\min}$ is the unscaled temperature of the coldest state at that coupling, and is the same state whose scaled temperature appears as $\tau_{\min}$ in table 4; the ratio column gives the span of the up-to-twelve-point window, which is the lever arm of the fit. Each shift is the largest absolute difference from the production central value over both windows and both coordinates. These two columns are part of the sensitivity envelope of table 4, not a separate estimate.

times, each time with one of the six fitted states removed, keeping the model fixed so that only one thing changes. Five points still over-determine a cubic. The largest resulting displacement of the intercept over all couplings is 1.09×10^{-5} , and the largest residual the production fit leaves on its own points is 1.77×10^{-6} ; both are smaller than every sensitivity in table 4. Neither is a bound on a bias shared by the six states, which is what a systematic error in the underlying charges would be.

The uncertainty budget. Table 4 carries a sensitivity for Ψ_{ext} alone, but y , μ , j and σ are extrapolated too, and the mass-curve check of (19) depends on y and μ directly. Table 7 collects what is recorded for each. The spreads for y , μ and σ are those of their own intercepts across the original degree checks; the spread for j is obtained by forming j from each pair of extrapolated (M, J) intercepts, which propagates their correlation rather than adding them in quadrature. The last two columns are the fit diagnostics of the preceding paragraph.

These are not a combined error bound and are not to be added. The effects they do not separate are radial resolution, which is measured at only 3 couplings (table 9); the asymptotic tail extraction, whose recorded spread is 1.31×10^{-8} and which can be biased without spreading; the nonlinear solver tolerance; the response linear solve; and the interpolation used in (19). Read as one number per row, the Ψ_{ext} sensitivity is the largest of the effects we can measure and not a bound on their sum.

Radial resolution. Production walks use $N = 64$, raised to 96 when a state is refused. To bound what that choice costs, the complete ascending walk and the complete $T = 0$ extrapolation were repeated at $N = 48$ and 56 at every one of the 13 couplings: 26 independent sequences, each using the production start spin, the production acceptance gates and the production escalation rule, so that the base resolution is the only thing varied. Only the ascending branch is walked, since the twelve coldest states that enter the fit all lie on it. Table 8 gives the outcome observable by observable, because a small change in Ψ_{ext} does not by itself bound a change in M , J , S , y or μ , and (19) uses the last two.

Across all 26 sequences the extremal intercept moves by at most 6.89×10^{-5} in Ψ_{ext} , 1.68×10^{-5}

α	δy	$\delta \mu$	δj	$\delta \sigma$	$\delta \Psi_{\text{ext}}$	fit residual	drop-one shift
0	0	1.33×10^{-7}	8.25×10^{-9}	3.10×10^{-7}	1.15×10^{-2}	8.98×10^{-7}	1.10×10^{-6}
0.005	1.23×10^{-6}	1.26×10^{-3}	8.32×10^{-5}	3.38×10^{-3}	1.12×10^{-1}	1.77×10^{-6}	1.09×10^{-5}
0.01	1.52×10^{-6}	7.69×10^{-4}	5.74×10^{-5}	2.23×10^{-3}	8.81×10^{-2}	1.54×10^{-6}	1.08×10^{-5}
0.02	5.38×10^{-6}	1.61×10^{-3}	2.93×10^{-4}	4.03×10^{-3}	1.60×10^{-1}	7.71×10^{-7}	8.50×10^{-6}
0.035	4.02×10^{-6}	8.43×10^{-4}	5.16×10^{-4}	1.63×10^{-3}	1.32×10^{-1}	5.48×10^{-7}	2.38×10^{-6}
0.05	4.25×10^{-8}	5.68×10^{-6}	3.27×10^{-6}	1.84×10^{-5}	1.86×10^{-2}	7.39×10^{-8}	3.98×10^{-7}
0.075	5.55×10^{-8}	9.14×10^{-6}	4.64×10^{-6}	4.35×10^{-5}	2.20×10^{-2}	7.29×10^{-8}	7.93×10^{-7}
0.1	3.52×10^{-7}	3.69×10^{-5}	1.78×10^{-5}	2.15×10^{-4}	2.10×10^{-2}	3.17×10^{-7}	1.35×10^{-6}
0.15	1.68×10^{-7}	1.08×10^{-5}	3.18×10^{-6}	8.02×10^{-5}	1.30×10^{-2}	3.79×10^{-8}	2.06×10^{-7}
0.2	1.62×10^{-7}	8.76×10^{-6}	2.59×10^{-6}	7.47×10^{-5}	1.12×10^{-2}	3.64×10^{-7}	1.21×10^{-6}
0.3	8.36×10^{-8}	4.46×10^{-6}	1.37×10^{-6}	4.72×10^{-5}	6.16×10^{-3}	2.32×10^{-7}	1.09×10^{-6}
0.4	2.73×10^{-7}	4.55×10^{-6}	1.64×10^{-6}	6.27×10^{-5}	6.96×10^{-3}	4.34×10^{-7}	4.70×10^{-7}
0.5	1.10×10^{-7}	1.09×10^{-6}	3.69×10^{-7}	1.81×10^{-5}	1.39×10^{-3}	3.85×10^{-7}	5.28×10^{-7}

Table 7. What is recorded for every extrapolated column of table 4, not only for Ψ_{ext} . The first four are intercept spreads across the original degree checks, δj propagated through the extrapolated M and J together; $\delta \Psi_{\text{ext}}$ is the envelope of table 4. The last two columns are the largest residual the production cubic leaves on its own six states and the largest displacement of its intercept when one of those states is removed; they are not independent contributions to be combined.

in μ and 5.59×10^{-7} in y , and the extrapolated charges by at most 5.62×10^{-5} , 8.13×10^{-5} and 7.09×10^{-5} relative in M , J and S respectively; the two remaining columns of table 7 move by at most 1.03×10^{-5} in j and 1.07×10^{-4} in σ . The quantity this has to be compared with is the fit sensitivity already quoted at the same coupling. Figure 7 draws the two together: at 26 of the 26 sequences the resolution difference is below that sensitivity, the worst ratio between them being 7.82×10^{-4} . Radial resolution is therefore not the leading term in the budget of table 7; the extrapolation is, by several orders of magnitude. Raising the coarse resolution from $N = 48$ to $N = 56$ reduces the difference in Ψ_{ext} at 9 of the 13 couplings.

Three things keep this short of a convergence proof. First, the escalation rule means the coldest states can be accepted at $N = 96$ on both sides; that happens at 20 of the 26 sequences, where the comparison is weaker than the two base resolutions suggest. Second, a coarse walk need not reach as cold as the production one, so the two fits can extrapolate from different distances and the difference between them is not purely a resolution effect; refitting both sides over a common temperature range isolates the resolution and leaves a worst difference 5.43×10^{-5} over the 26 sequences in which both sides keep a full window. Third, the shipped measurement was produced under Python 3.11.9, NumPy 2.4.6, SciPy 1.17.1 and this study runs under Python 3.12.3, NumPy 1.26.4, SciPy 1.11.4, so each difference above bounds the change of resolution and the change of environment together. Re-walking at the production resolution here, at $\alpha = 0.02, 0.15, 0.5$, attributes at most 1.13×10^{-5} to the environment alone. That is not a negligible share: at two of the three control couplings it exceeds at least one of the two resolution differences measured at the same coupling, and at $\alpha = 1/2$ it exceeds both. The entries of table 8 are therefore upper bounds on the effect of the resolution, not estimates of it, and the true resolution sensitivity of this calculation is smaller than the table says. Table 9 is free of that caveat: its two sides were both computed in the shipped environment, at the 3 couplings where that comparison exists, and there the extrapolated potential moves by at most 2.56×10^{-5} and μ by at most 6.03×10^{-6} .

The integral comparison in (19) has its own interpolation error and correlations with both sets of estimates; we report its discrepancy separately instead of interpreting it as coverage by independent error bars.

The profile atlas. Figure 1 is produced by a separate run, whose only purpose is to exhibit the solutions behind the rest of the paper and to test them along a second route. It fixes $q = 0.6$,

α	N	$\delta M/M$	$\delta J/J$	$\delta S/S$	$\delta \Psi_{\text{ext}}$	δy	$\delta \mu$
0	48	1.10×10^{-11}	7.53×10^{-13}	1.77×10^{-14}	2.78×10^{-6}	0	2.53×10^{-11}
0	56	2.82×10^{-11}	4.59×10^{-13}	6.11×10^{-14}	5.24×10^{-6}	0	6.13×10^{-11}
0.005	48	2.18×10^{-5}	3.13×10^{-5}	2.80×10^{-5}	1.13×10^{-5}	1.00×10^{-7}	2.04×10^{-6}
0.005	56	2.18×10^{-5}	3.13×10^{-5}	2.80×10^{-5}	1.15×10^{-5}	1.00×10^{-7}	2.04×10^{-6}
0.01	48	5.62×10^{-5}	8.13×10^{-5}	7.09×10^{-5}	6.89×10^{-5}	5.33×10^{-7}	4.57×10^{-6}
0.01	56	2.46×10^{-5}	3.71×10^{-5}	3.18×10^{-5}	2.62×10^{-5}	2.43×10^{-7}	3.51×10^{-7}
0.02	48	1.37×10^{-5}	1.61×10^{-5}	1.69×10^{-5}	2.84×10^{-5}	2.23×10^{-7}	6.69×10^{-6}
0.02	56	6.44×10^{-7}	9.05×10^{-7}	8.04×10^{-7}	7.90×10^{-7}	1.25×10^{-8}	9.20×10^{-8}
0.035	48	5.43×10^{-9}	1.10×10^{-5}	6.64×10^{-7}	7.83×10^{-6}	2.82×10^{-7}	1.68×10^{-5}
0.035	56	1.40×10^{-7}	3.01×10^{-6}	3.97×10^{-7}	2.48×10^{-6}	7.72×10^{-8}	4.93×10^{-6}
0.05	48	1.98×10^{-7}	3.09×10^{-7}	2.44×10^{-7}	1.03×10^{-7}	1.17×10^{-8}	1.96×10^{-8}
0.05	56	5.46×10^{-7}	8.48×10^{-7}	6.61×10^{-7}	1.22×10^{-6}	3.22×10^{-8}	4.47×10^{-8}
0.075	48	1.47×10^{-6}	2.30×10^{-6}	1.67×10^{-6}	1.45×10^{-6}	1.35×10^{-7}	1.44×10^{-7}
0.075	56	3.31×10^{-11}	2.02×10^{-12}	1.24×10^{-13}	9.15×10^{-8}	1.18×10^{-13}	8.18×10^{-11}
0.1	48	1.41×10^{-6}	2.24×10^{-6}	1.57×10^{-6}	1.24×10^{-6}	1.76×10^{-7}	1.95×10^{-7}
0.1	56	2.35×10^{-7}	3.72×10^{-7}	2.59×10^{-7}	1.73×10^{-7}	2.92×10^{-8}	3.16×10^{-8}
0.15	48	2.98×10^{-6}	4.79×10^{-6}	3.17×10^{-6}	7.69×10^{-7}	5.59×10^{-7}	5.16×10^{-7}
0.15	56	2.98×10^{-6}	4.79×10^{-6}	3.17×10^{-6}	7.61×10^{-7}	5.59×10^{-7}	5.16×10^{-7}
0.2	48	4.38×10^{-7}	7.30×10^{-7}	4.81×10^{-7}	6.68×10^{-7}	1.12×10^{-7}	1.24×10^{-7}
0.2	56	1.31×10^{-7}	2.19×10^{-7}	1.54×10^{-7}	4.73×10^{-7}	3.35×10^{-8}	3.78×10^{-8}
0.3	48	2.17×10^{-8}	3.84×10^{-8}	2.51×10^{-8}	4.71×10^{-8}	8.43×10^{-9}	1.04×10^{-8}
0.3	56	1.84×10^{-12}	2.73×10^{-12}	5.19×10^{-14}	6.96×10^{-9}	6.00×10^{-13}	5.06×10^{-14}
0.4	48	6.88×10^{-8}	1.29×10^{-7}	9.42×10^{-8}	1.20×10^{-7}	3.64×10^{-8}	4.79×10^{-8}
0.4	56	1.59×10^{-11}	2.11×10^{-12}	6.81×10^{-13}	2.09×10^{-9}	5.94×10^{-13}	4.00×10^{-11}
0.5	48	1.09×10^{-9}	2.14×10^{-9}	2.02×10^{-9}	3.92×10^{-8}	7.31×10^{-10}	9.91×10^{-10}
0.5	56	1.12×10^{-9}	2.23×10^{-9}	1.35×10^{-8}	5.93×10^{-7}	7.59×10^{-10}	1.06×10^{-9}

Table 8. The radial-resolution study: the complete ascending walk and the complete $T = 0$ extrapolation, repeated at two coarser base resolutions at every coupling. Each entry is the difference between the production value at $N = 64$ and the value the coarser sequence extrapolates to; the two sequences are independent from the first continuation step onwards. M , J and S are quoted relative and the rest absolute, matching how each is used in the text. The columns are separate because they are separate: none of them bounds another.

α	Ψ_{ext} at $N = 64$	change at $N = 48$	μ at $N = 64$	change	change in y
0.02	2.472827	2.56×10^{-5}	2.255058	6.03×10^{-6}	2.01×10^{-7}
0.1	1.166214	7.49×10^{-7}	2.410729	1.47×10^{-7}	1.36×10^{-7}
0.5	1.444398	1.26×10^{-7}	2.888535	4.31×10^{-11}	9.85×10^{-11}

Table 9. The resolution comparison recorded with the shipped measurement, at the 3 couplings where it exists. It carries less coverage than table 8 and one advantage over it: both sides were computed in the same environment, so these differences are attributable to the resolution alone.

climbs the spin from the vacuum in steps of that size, and then climbs the coupling at fixed spin, seeding each Newton solve with the previous solution and using no response predictor. The production walks do the opposite: they climb the coupling first, with a predictor, and then descend or ascend in spin. The two routes visit the same 13 solutions and agree on M , J , S and T to 2.51×10^{-11} relative. Each state in the atlas passes the same boundary and independent tensor-residual gates as the production states, and the run also checks that the vacuum member reproduces the closed-form Myers–Perry horizon squashing $1/(1 - q^2)$ and that $w(r_H) = \Omega_H$ is recovered to machine precision.

Data and code. The supplementary table contains the 356 accepted states, their observables and acceptance diagnostics, one row per solution, so that every point of figure 2 and every row of

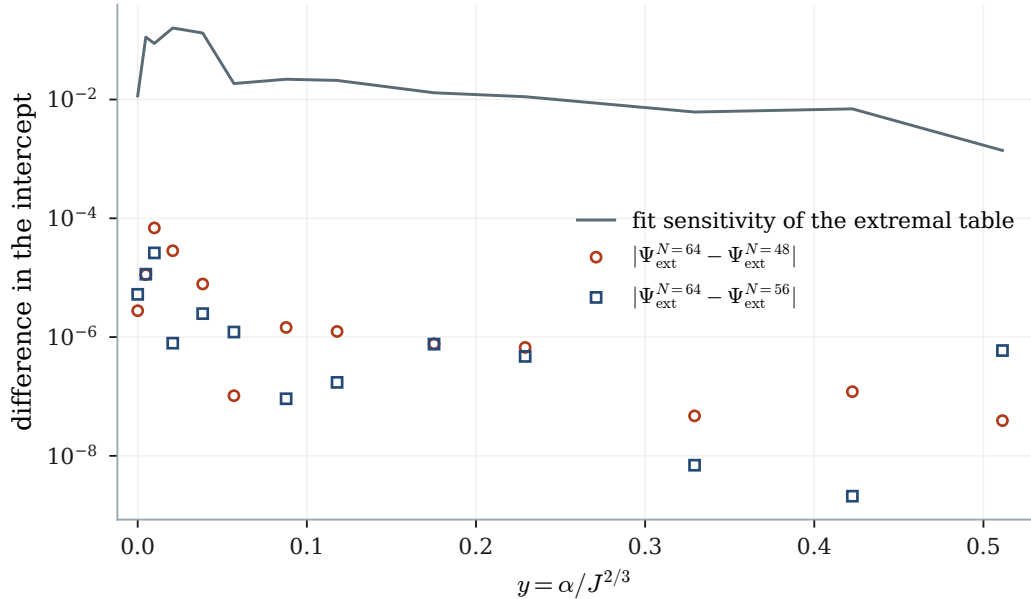


Figure 7. The resolution study against the quantity it has to be compared with. Markers are the change in the extremal intercept when the whole sequence is recomputed at a coarser base resolution; the line is the fit sensitivity already quoted for the same coupling in table 4. The gap between them is the statement: at every coupling the extrapolation model dominates the radial resolution. The vertical scale is logarithmic and the two marker sets are the two coarse resolutions of table 8.

tables 3 and 4 can be traced to the state it came from. The radial amplitudes of figure 1 are stored separately, on the common grid used to draw them, and the fitted points, residuals, competing intercepts and drop-one shifts behind figure 3 and tables 6 and 7 are exported with them. The radial-resolution study is a separate artifact carrying every state of all 26 coarse sequences, so that table 8 and figure 7 can be recomputed without repeating the solves. The accompanying source tree contains the solver, symbolic derivation scripts, tests and figure-generation script. The latter also exports the fitted intercepts and sensitivity envelopes used in this version, together with source hashes.

Every number in this manuscript is generated from the recorded measurement by that script and none is typed into the source, so the PDF, the figures, the tables and the data cannot disagree with one another: rebuilding reproduces them or fails. What this arrangement does not by itself provide is an immutable identifier for the state it was built from. These materials document the calculation; this manuscript does not claim an existing public archival DOI.

B The near-horizon entropy function

The comparison in section 5 uses the extremal entropy obtained from the near-horizon geometry, following section 4.2 of [11]. We summarise our rederivation, which is independent of that reference in execution though not in content.

For equal angular momenta the extremal near-horizon geometry is homogeneous, with isometry group $SO(2,1) \times SU(2) \times U(1)$; see [20] for the general classification of such geometries. It can be written in terms of the $SU(2)$ invariant one-forms σ_i as

$$ds^2 = v_1 \left(-r^2 dt^2 + \frac{dr^2}{r^2} \right) + \frac{v_2}{4} (\sigma_1^2 + \sigma_2^2) + \frac{v_3}{4} (\sigma_3 + k r dt)^2, \quad (24)$$

with $v_{1,2,3}$ and k constants. On (24) both R and $\mathcal{L}_{GB}$ are constant, and Sen's entropy function

[12] reduces the problem to algebra: with

$$f(v, k) = \int d\theta d\varphi d\psi \sqrt{-g} \frac{R + \alpha \mathcal{L}_{\text{GB}}}{16\pi} = \frac{\pi}{8} v_1 v_2 \sqrt{v_3} (R + \alpha \mathcal{L}_{\text{GB}}), \quad (25)$$

the equations of motion are $\partial f / \partial v_i = 0$, the angular momentum is $J = \partial f / \partial k$, and the entropy is the Legendre transform $S_{\text{ext}} = 2\pi(kJ - f)$ evaluated at the extremum.

We derived R and $\mathcal{L}_{\text{GB}}$ on (24) symbolically and checked them against the same numerical curvature evaluator used for acceptance; we then verified that the extremal solution satisfies the full field equations $G_{\mu\nu} + \alpha H_{\mu\nu} = 0$ component by component, since extremising a reduced functional and solving the equations are not the same statement. At $\alpha = 0$ the system solves exactly, $k = 1$, $v_2 = 4v_1$, $v_3 = 8v_1$, giving $S_{\text{ext}} = 2\pi J$, which is extremal Myers–Perry ($S = 2\pi^2 a^3$, $J = \pi a^3$ at $r_+ = a$) with nothing tuned: the $1/16\pi G$ of the action and the 2π of the Legendre transform were both fixed beforehand. At finite coupling the branch is a root of an explicit polynomial system, and it reproduces eqs. (4.11) and (4.12) of [11] to the precision quoted in section 5.

Under $v_i \rightarrow \lambda v_i$, $\alpha \rightarrow \lambda \alpha$ both J and S_{ext} scale as $\lambda^{3/2}$, so $\sigma = S_{\text{ext}}/J$ and $y = \alpha/J^{2/3}$ are invariant and are what figure 6 compares. We emphasise that this appendix produces no mass: the near-horizon geometry of an extremal black hole determines its entropy as a function of its charges and no more, a point made in [11] itself, where the absence of the corresponding $E(J)$ diagram is noted as an open problem.

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